ESCI 4170 Final Project

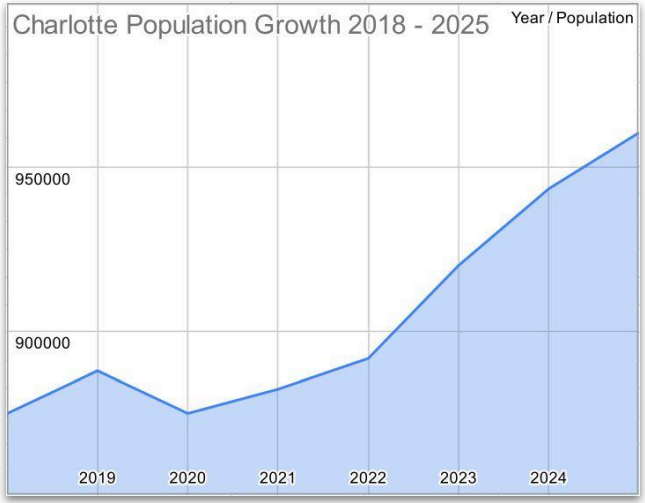
Urban Land Cover Change and Urbanization of Charlotte, NC

Christopher Horne and Jenna Bess

Sentinel-2 Geospatial Data 2018-2025

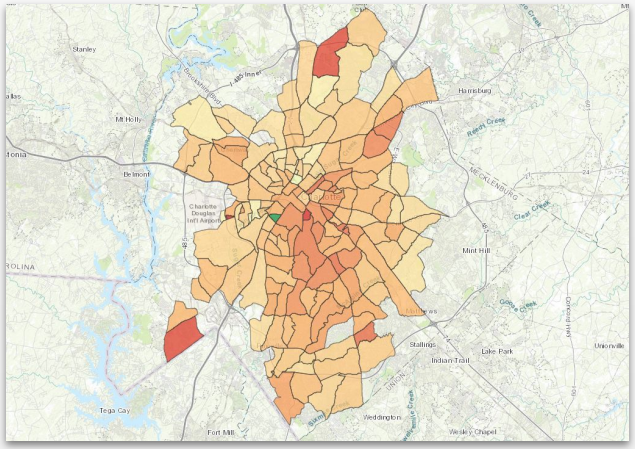
INTRODUCTION

Why Assess Land Cover and Urbanization in Charlotte?

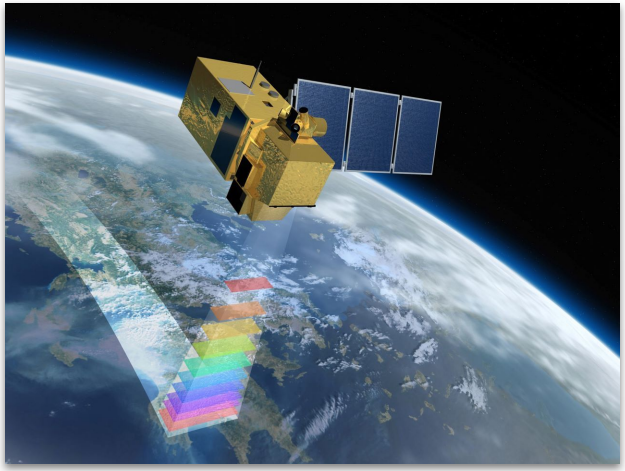


- Charlotte projected to reach 1 million residents by 2028
    - 6th largest population increase in the country in 2024
- Over 1,000 acres of canopy lost since 2018
- Canopy projected to fall to 40% by 2050
- Important for prevention of tree canopy loss

- Rapid development accelerating land transitions
- Increasing need to preserve green space and manage growth
- Analyze impact of new developments in the city

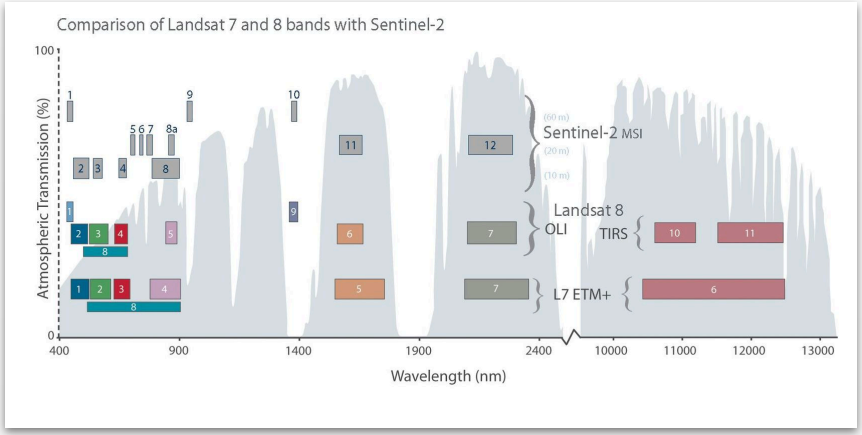


Why Sentinel-2?



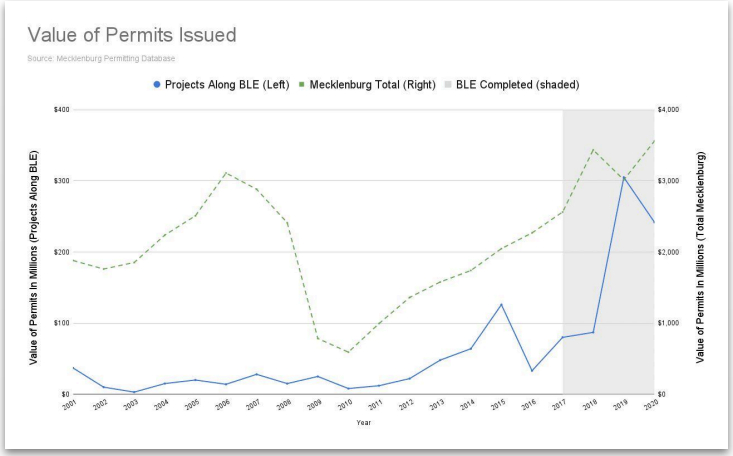
- 10 meter spatial resolution suitable for city-scale
- Frequent revisit time *[5 day cycle period]*
- NIR and SWIR bands
- Multi-year consistency

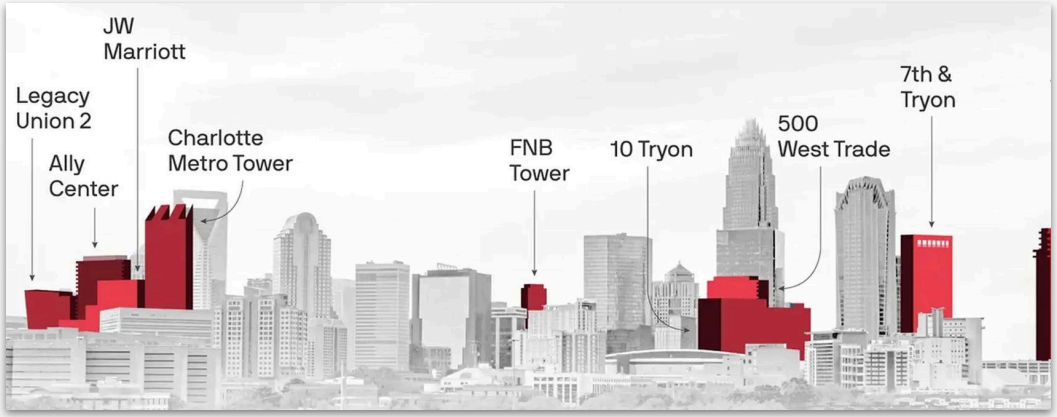
- Ideal for vegetation and urban analysis
- High spectral resolution
- Provides bands needed for thorough analysis
- Covers the Charlotte area well



Project Goals

- Map land cover in 2018 and 2025
- Analyze vegetation, moisture and urbanization changes
- Extract spectral and textural indicators
- Perform supervised classification and change detection

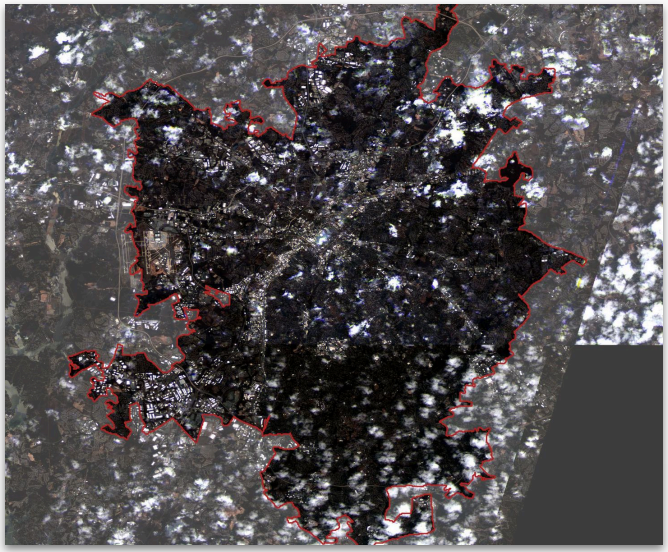




- Relate Results to Charlotte development trends
- Identify fastest growing regions

Challenges

- Cloud cover in several years causing data gaps
  - *Primarily used 2018 and 2025 to avoid cloud cover issues and avoid using high cloud cover all together*
- Spectral similarity in urban materials such as roads and buildings
  - *Combined these impervious surfaces into one classification*
- Differentiating trees and grass in MLC classification
  - *Increased the amount of samples*
- Bare soil versus impervious surface identification
  - *Took samples of bare soil in many unique areas*
- Limited ROI classification samples
  - *Took a consistent amount of samples for both years*
- Keeping multi-year data consistent
  - *Use July data for all years*
- Issues with GLCM data, lead to removal from analysis

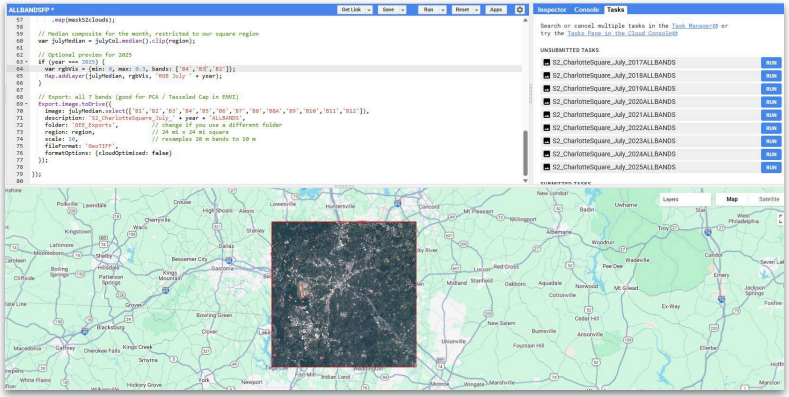


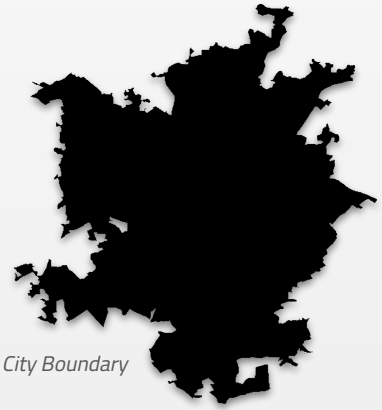
METHODS

Study Area & Data

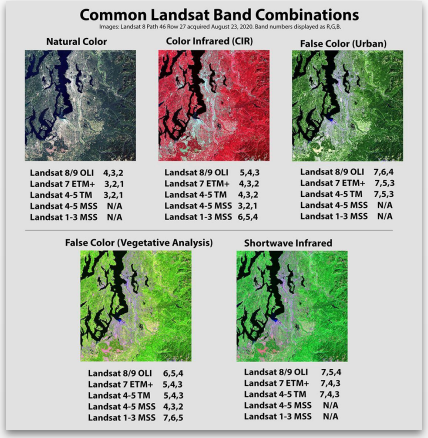
- Study Area: The City of Charlotte, North Carolina
- Rapidly urbanizing region with high development activity
- Sentinel-2 surface reflectance data
- Downloaded 12 multispectral bands from visible to SWIR
- July 2018 and 2025 used for comparison
- July 2018, 2020, 2024, 2025 used for analysis and NDVI Time Series
- Google Earth Engine, ENVI, and Photoshop
- Charlotte City Boundary
- National Land Cover Database for Analysis

Google Earth Engine View





Charlotte City Boundary

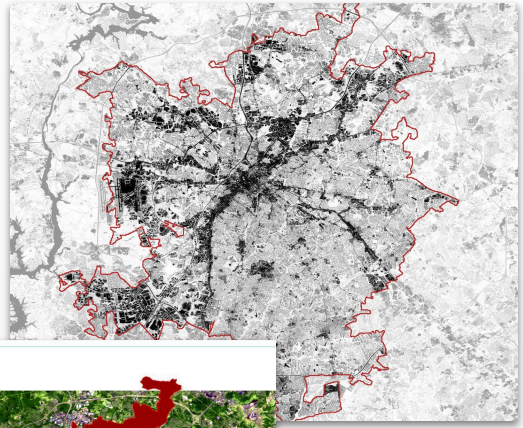


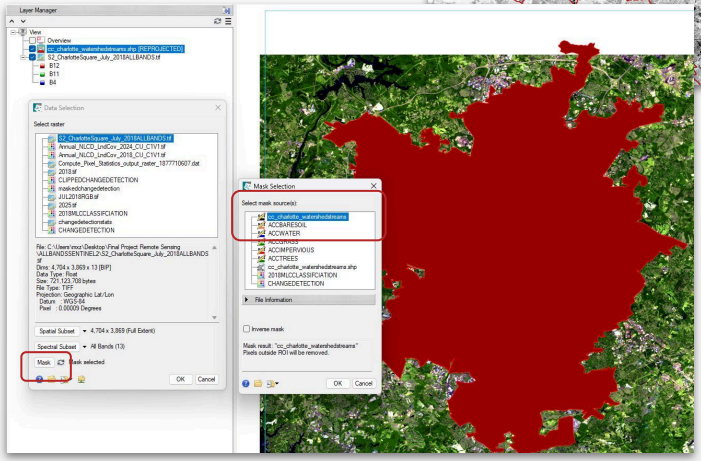
Landsat Band Combinations

Preprocessing and Data Retrieval

- Created a Google Earth Engine script to download Sentinel-2 Data for the region
- Used GEE to mask clouds with the QA60 band to filter for cloud free images
- Clipped data to Charlotte City Boundary using downloaded data
    - Clipped using masks in ENVI when running operations
    - Clipped in Photoshop for Final Viewing
- Consistent spatial extent used for all years
- Checked each for cloud cover/ abnormalities
- Used July images to prevent seasonal changes
- Filter for entire month of Sentinel-2 Data
    - July 1st - July 31st

Masking Result from Photoshop

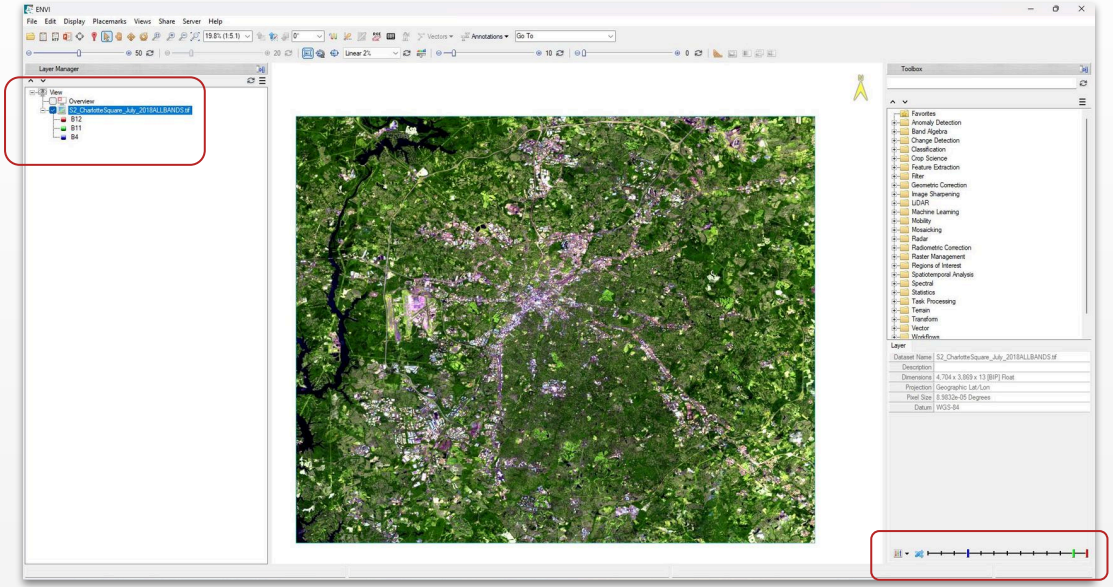




Masking Procedure in ENVI

Spectral Composites

- Urban False Color
  - Bands 12, 11, 4
- SWIR Urban Composite
  - Bands 12, 8, 4
- Standard Color Infrared Composite
  - Bands 8, 4, 3
- Soil Composite
  - Bands 12, 11, 2



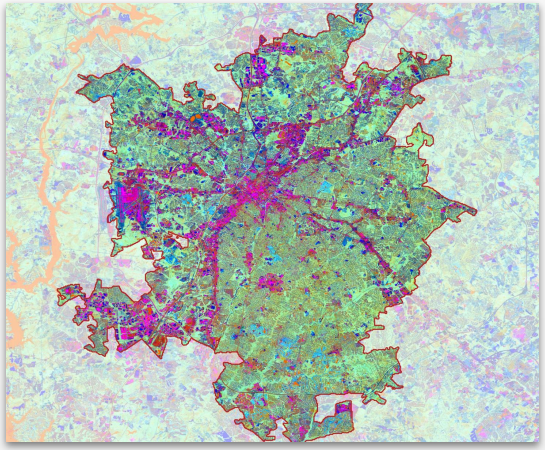
False Color Composite Method

- Each of the spectral composites highlight different land cover properties
- Used ENVI Data Manager to view these specific bands of Sentinel and exported them as TIF files

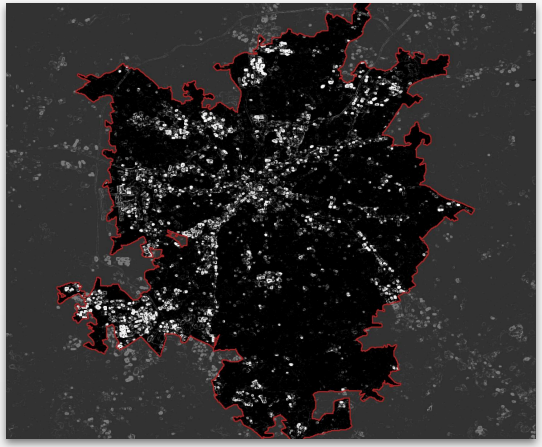
PCA and Texture Analysis

- Principal Component Analysis was used to highlight subtle spectral variance
- Helps to detect early development or canopy change
- Gray Level Co-Occurrence Matrix was performed, but excluded because of data errors
- GLCM metrics used: Contrast, Entropy and Homogeneity
- These metrics were chosen to show fragmentation, structure and randomness

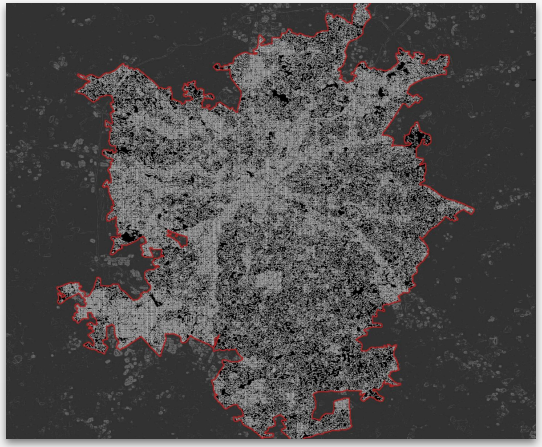
PCA 2025



GLCM CONTRAST (data errors)



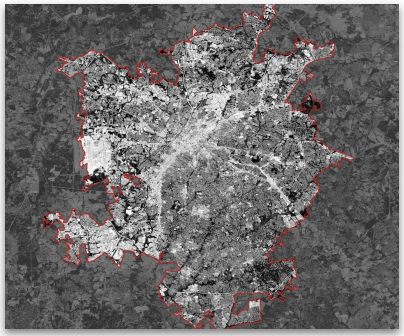
GLCM ENTROPY (data errors)



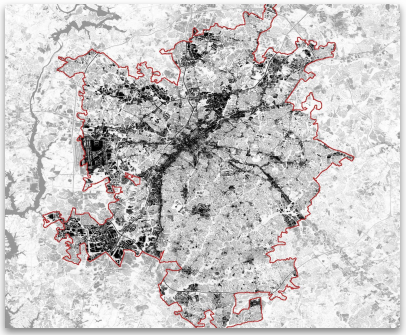
NDVI, NDMI, and NDBI

- NDVI
  - Normalized Difference Vegetation Index
    - Indicated Vegetation Loss
- NDMI
  - Normalized Difference Moisture Index
    - Detects Moisture Changes
- NDBI
  - Normalized Difference Built-Up Index
    - Highlights Urban Surfaces
- Band Math Formulas were used to calculate all the indexes

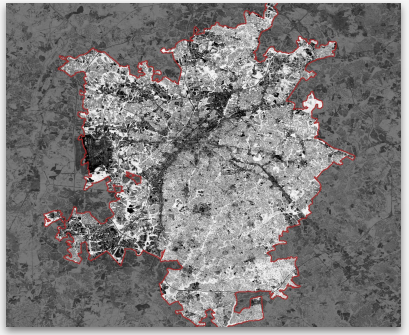
NDBI



NDVI



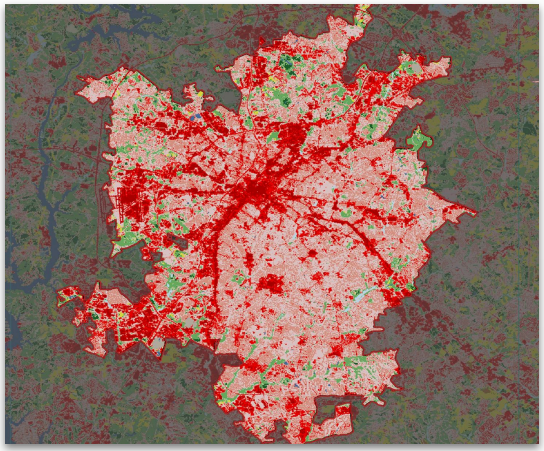
NDMI



Supervised Classification and Change Detection

- Maximum Likelihood Classification Used to Classify Land Cover Types
- Five Classes Identified
    - Trees
    - Grass
    - Impervious Surfaces
    - Water
    - Bare Soil

NLCD 2018

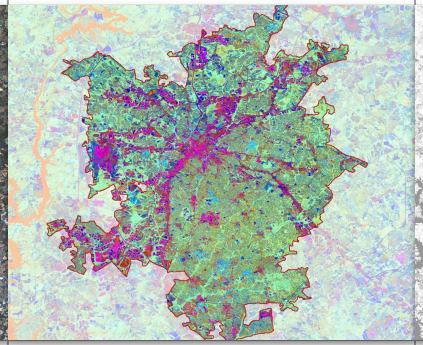
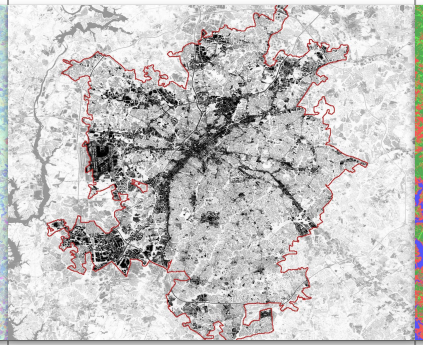
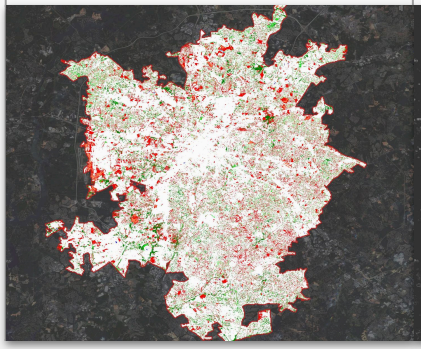
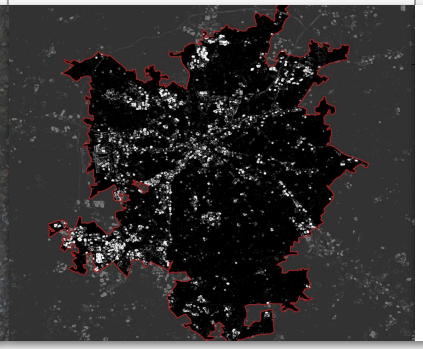
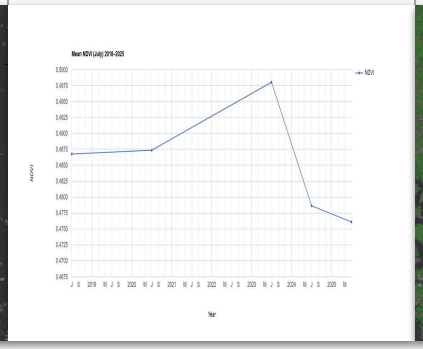
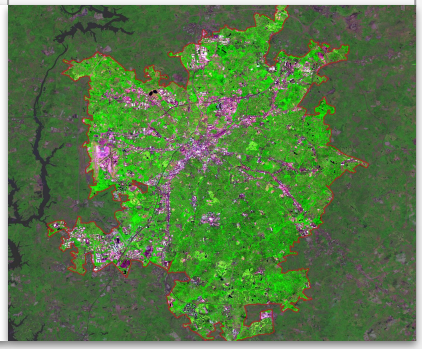


MLC 2018

- Consistent Quantity of ROI Training Samples
- Change Detection was performed using Thematic Change
- Identifies Class Changes and Development Characteristics

RESULTS

Overview of Products

RGB True Color Composite	Principal Component Analysis	Normalized Difference NDVI/ NDMI/ NDBI	Maximum Likelihood Classification
			
Change Detection using MLC	**Gray Level Co-Occurrence Matrix**	**NDVI Time Series Chart**	**False Color Composites**
			

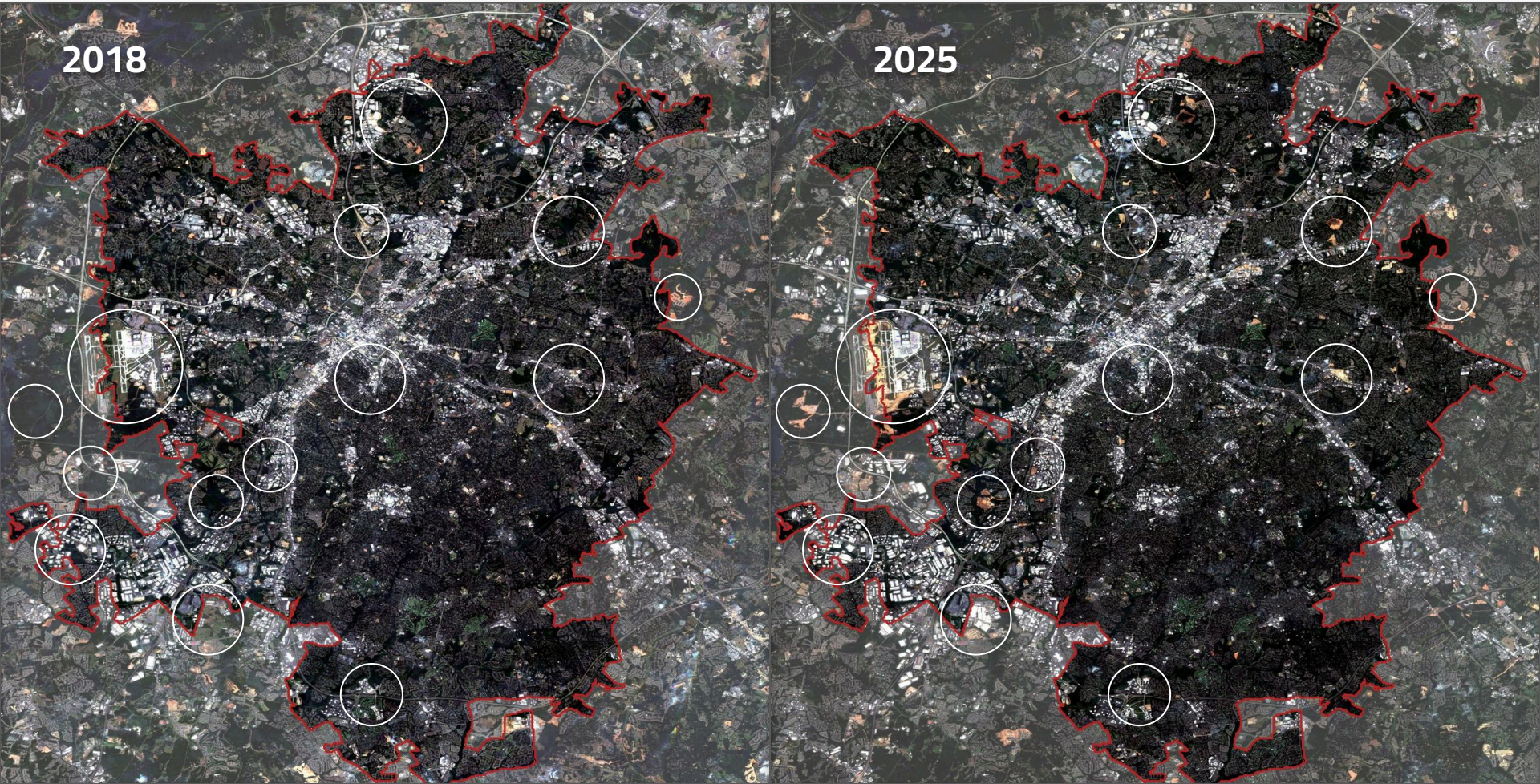
RGB Composite Charlotte NC *2018*

Sentinel-2 Data
Bands - 4, 3, 2
True Color Composite
Charlotte, NC Region with City Boundaries
Pulled with Google Earth Engine
July 2018

RGB Composite Charlotte NC *2025*

Sentinel-2 Data
Bands - 4, 3, 2
True Color Composite
Charlotte, NC Region with City Boundaries
Pulled with Google Earth Engine
July 2025

RGB Composite Charlotte NC *2018 vs 2025*

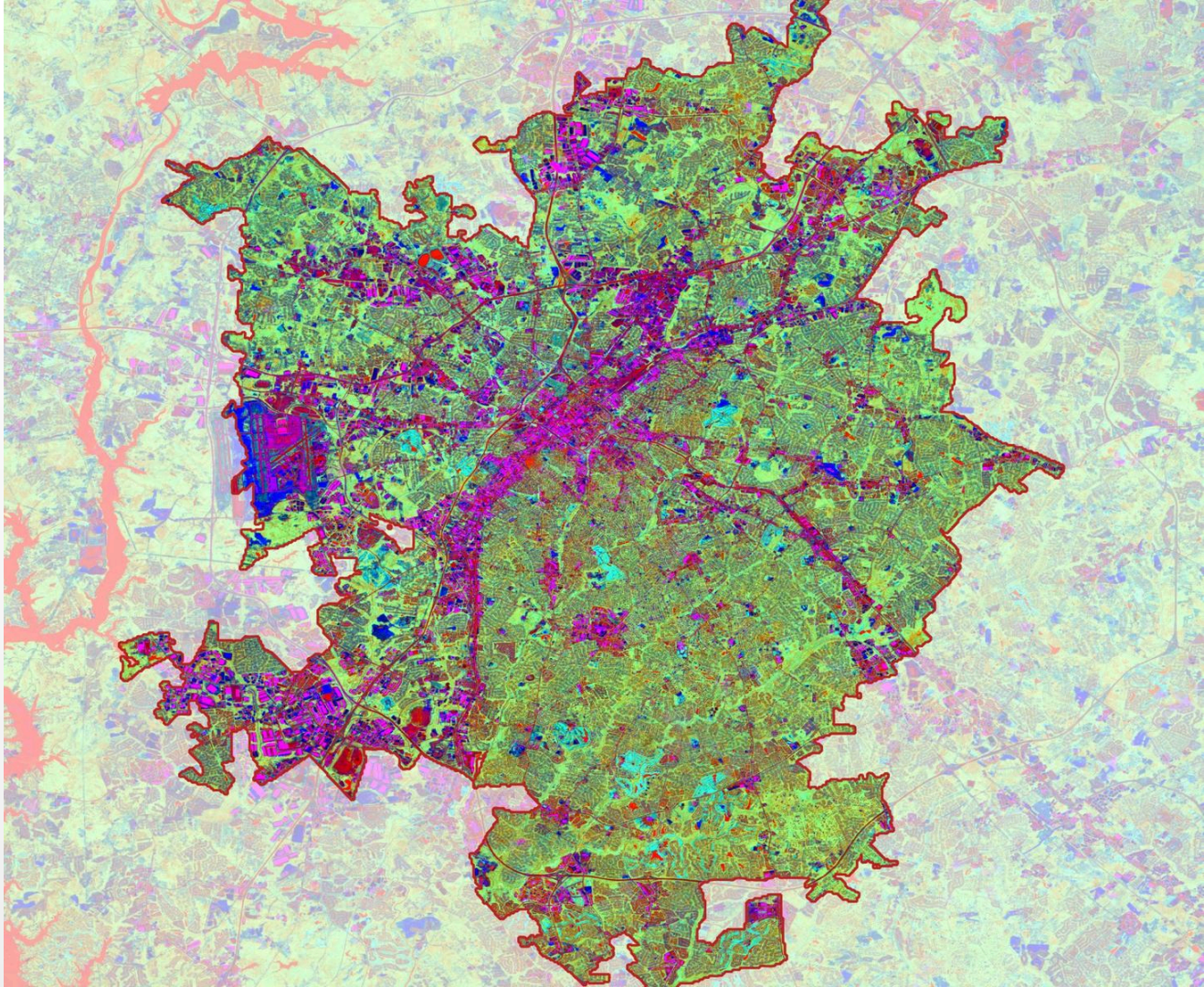


PCA
Charlotte NC
2018

PCA highlights spectral patterns shaped by vegetation dominance and early development

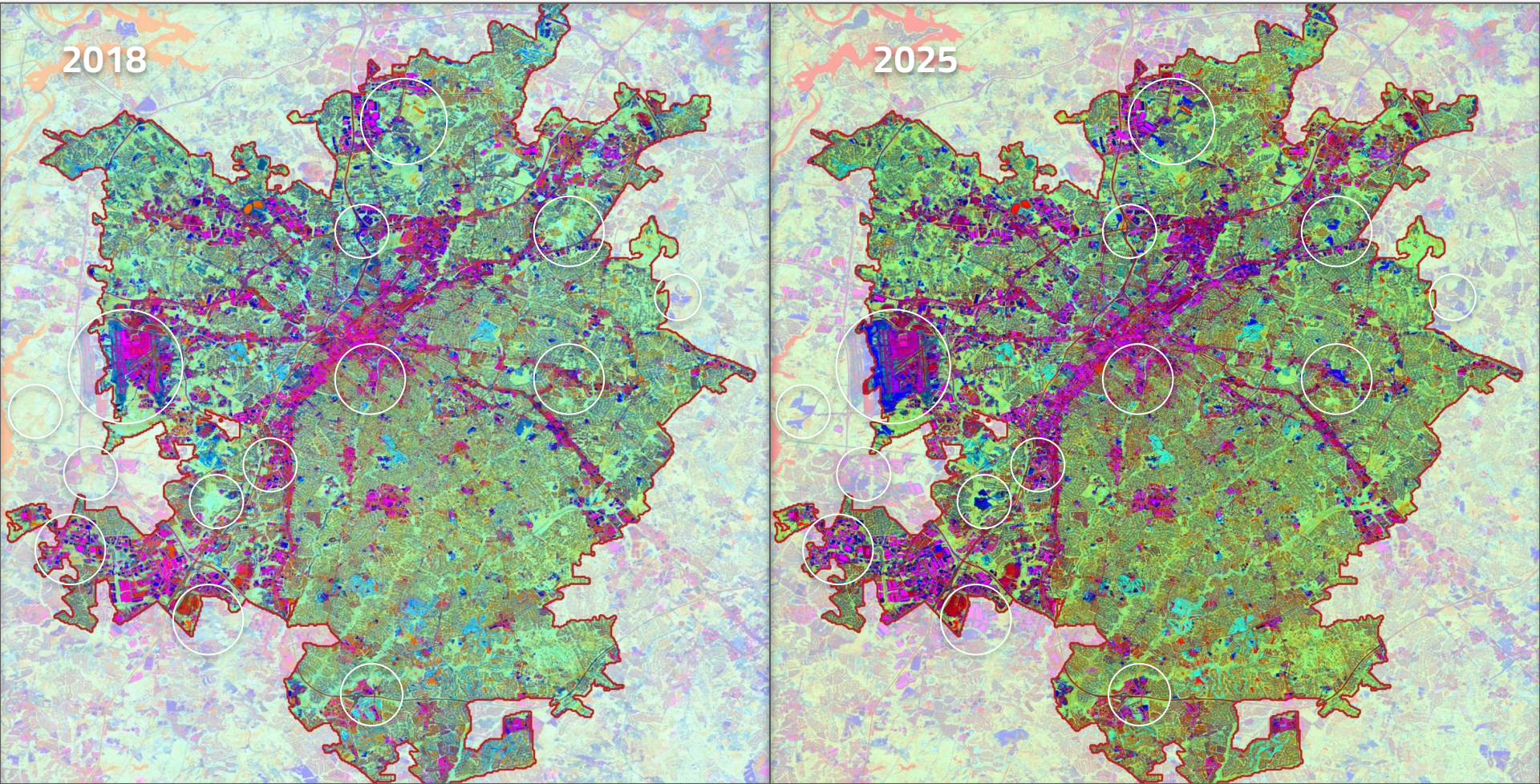
PCA
Charlotte NC
2025

Increased PCA variance reflects intensified development and altered land cover structure



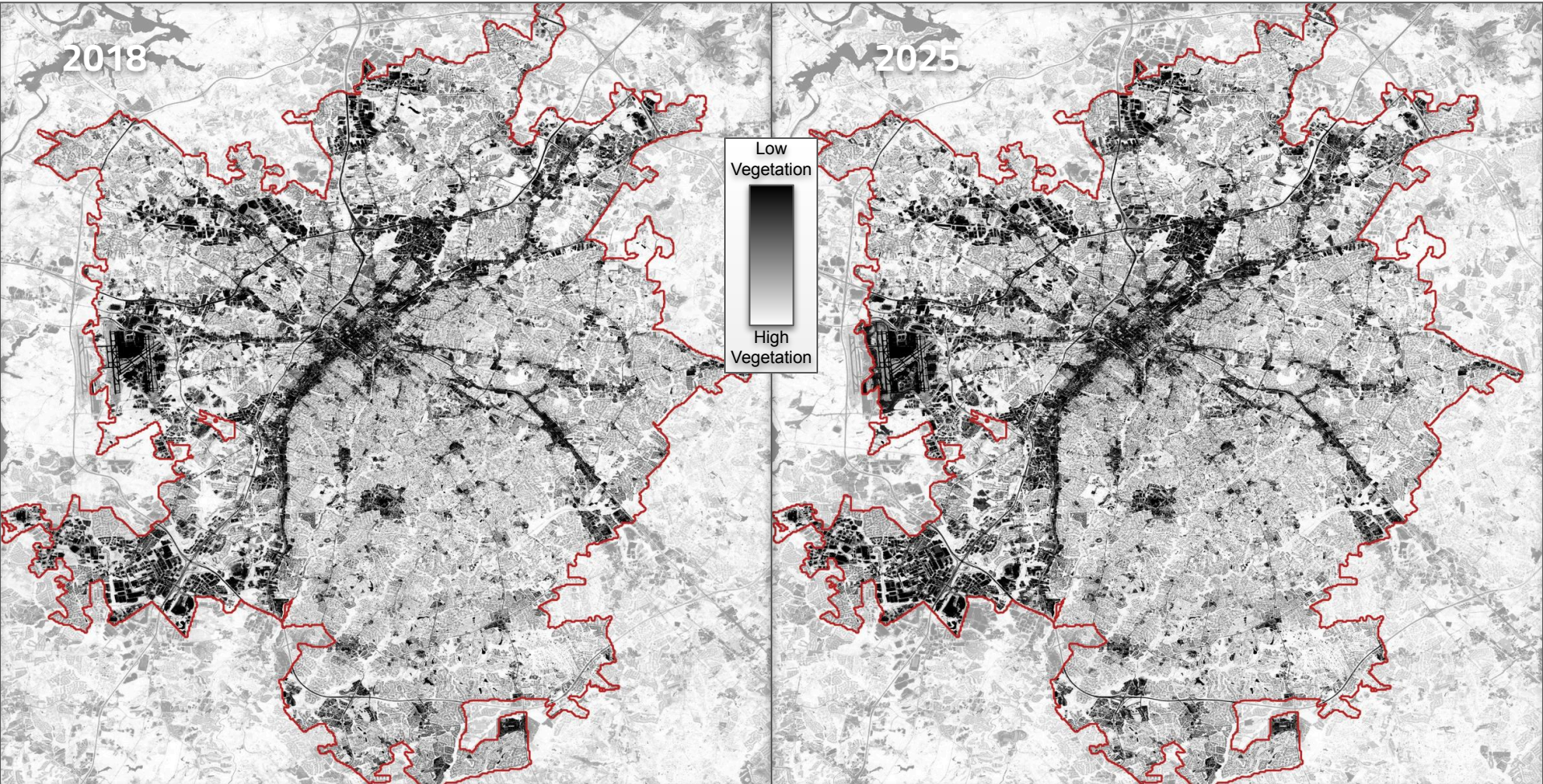
Principal Component Analysis (PCA) Charlotte NC *2018 vs 2025*

Shifts in PCA brightness reveal areas transformed by urban expansion and canopy loss

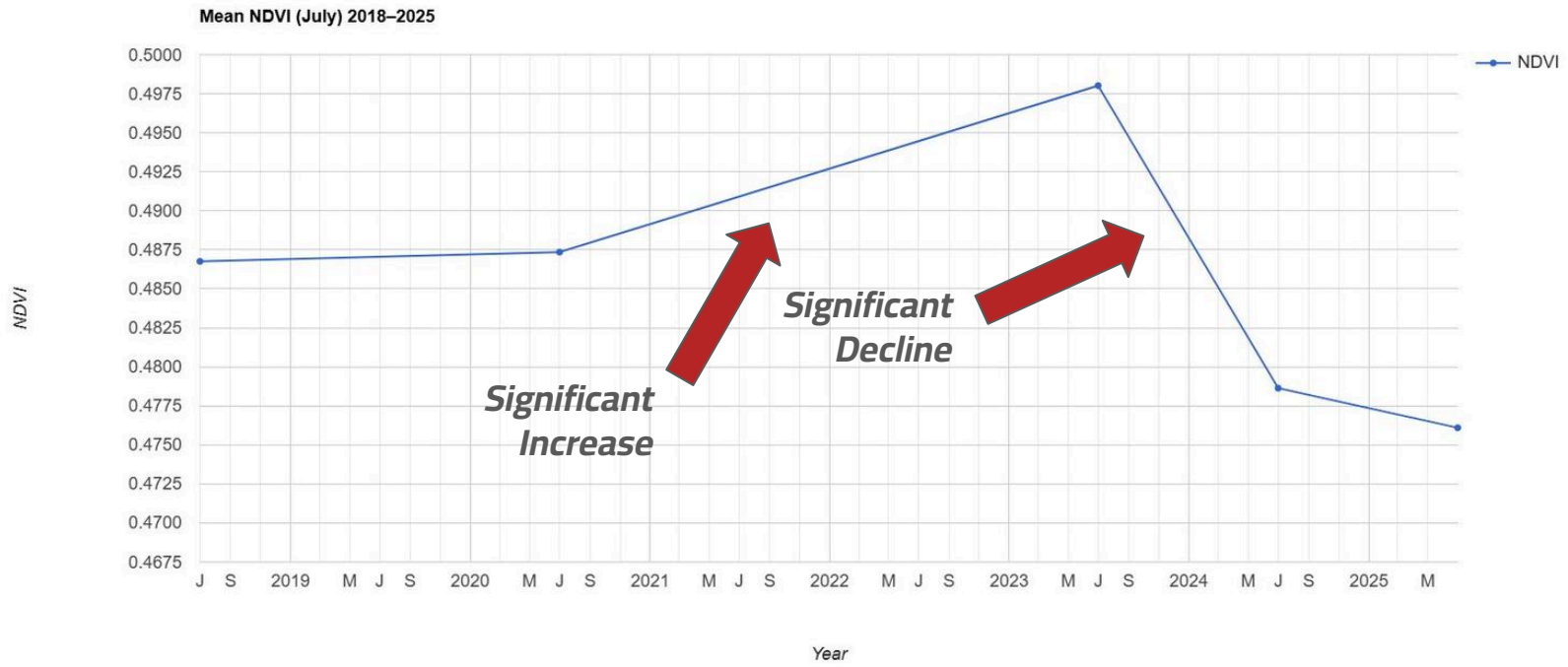


Normalized Difference Vegetation Index *2018 vs 2025*

Darkening NDVI tones indicate vegetation loss and increased land clearing



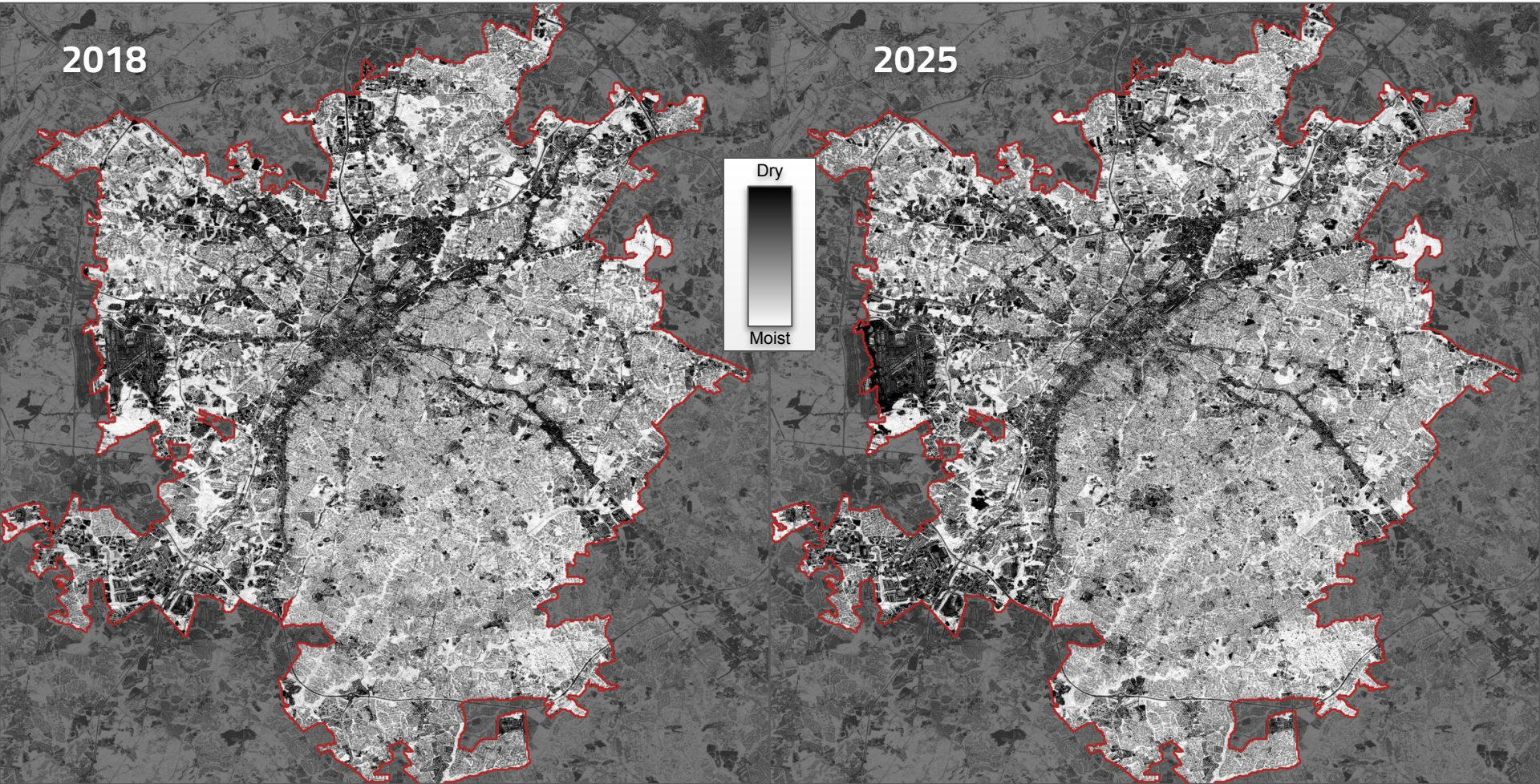
Normalized Difference Vegetation Index Time Series



Normalized Difference Vegetation Index (NDVI) highlighting vegetation cover across years

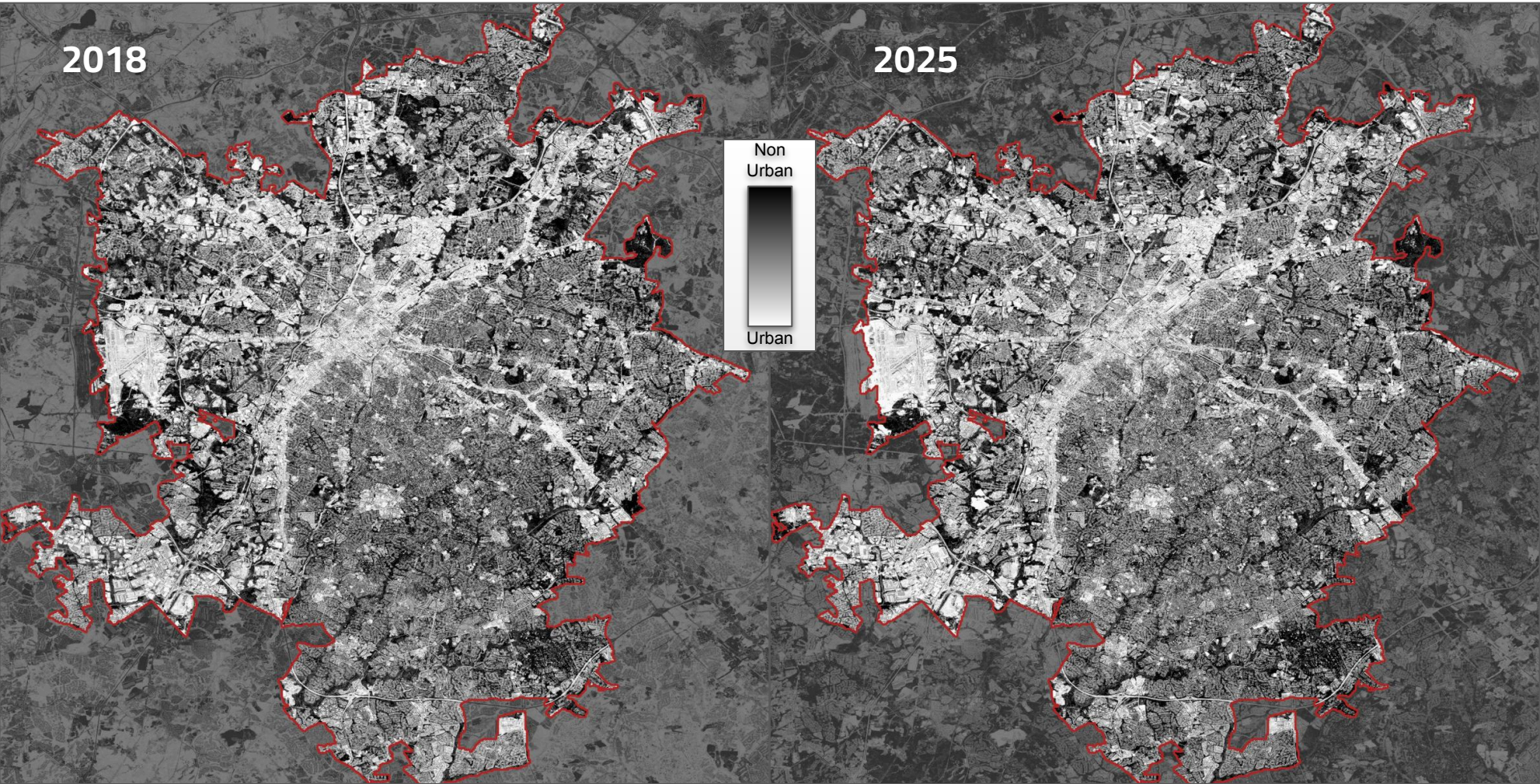
Normalized Difference Moisture Index *2018 vs 2025*

Lower NDMI values reflect canopy drying and soil disturbance associated with development



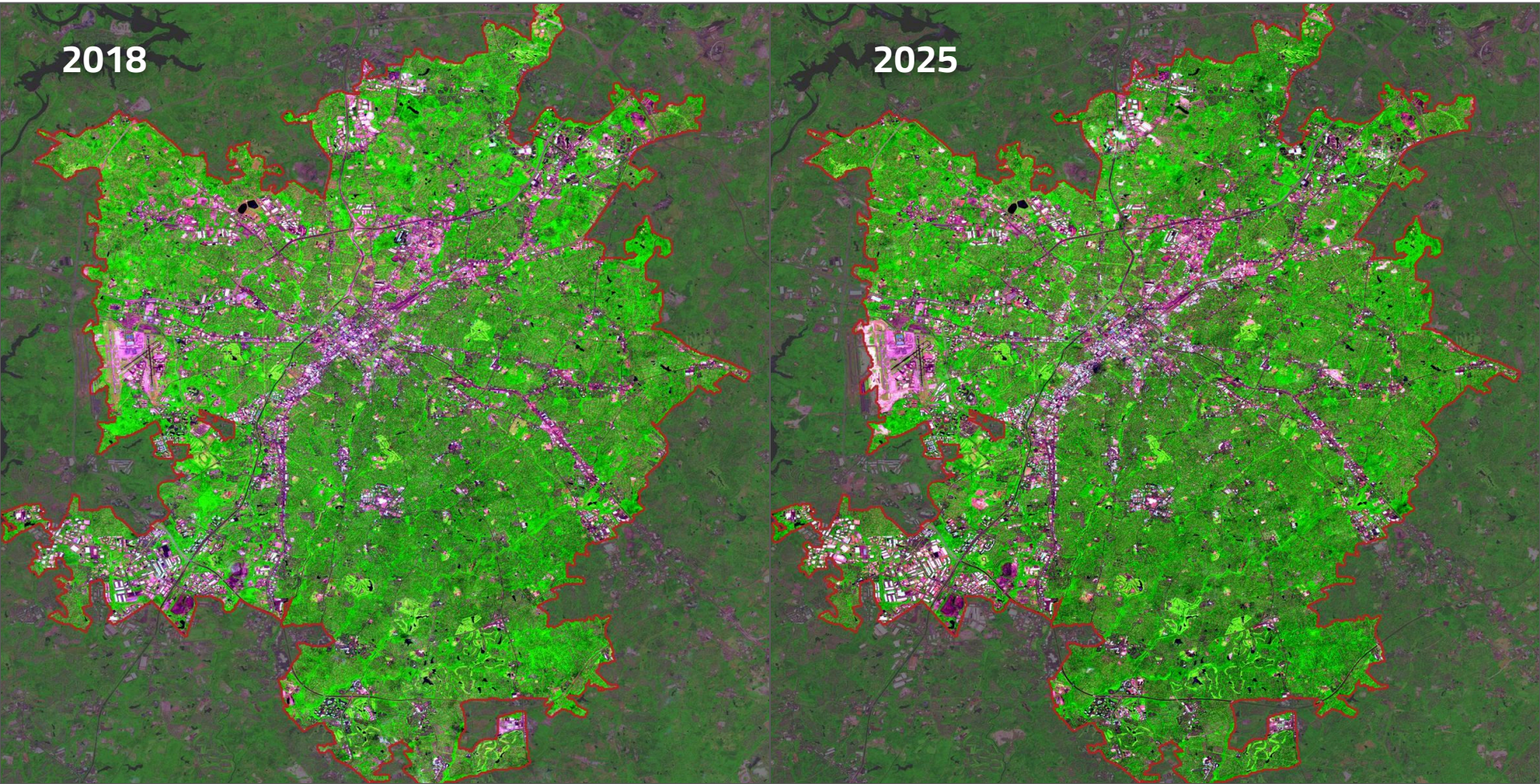
Normalized Difference Built-Up Index *2018 vs 2025*

Increasing NDBI brightness signals expanding impervious surfaces and new urban growth



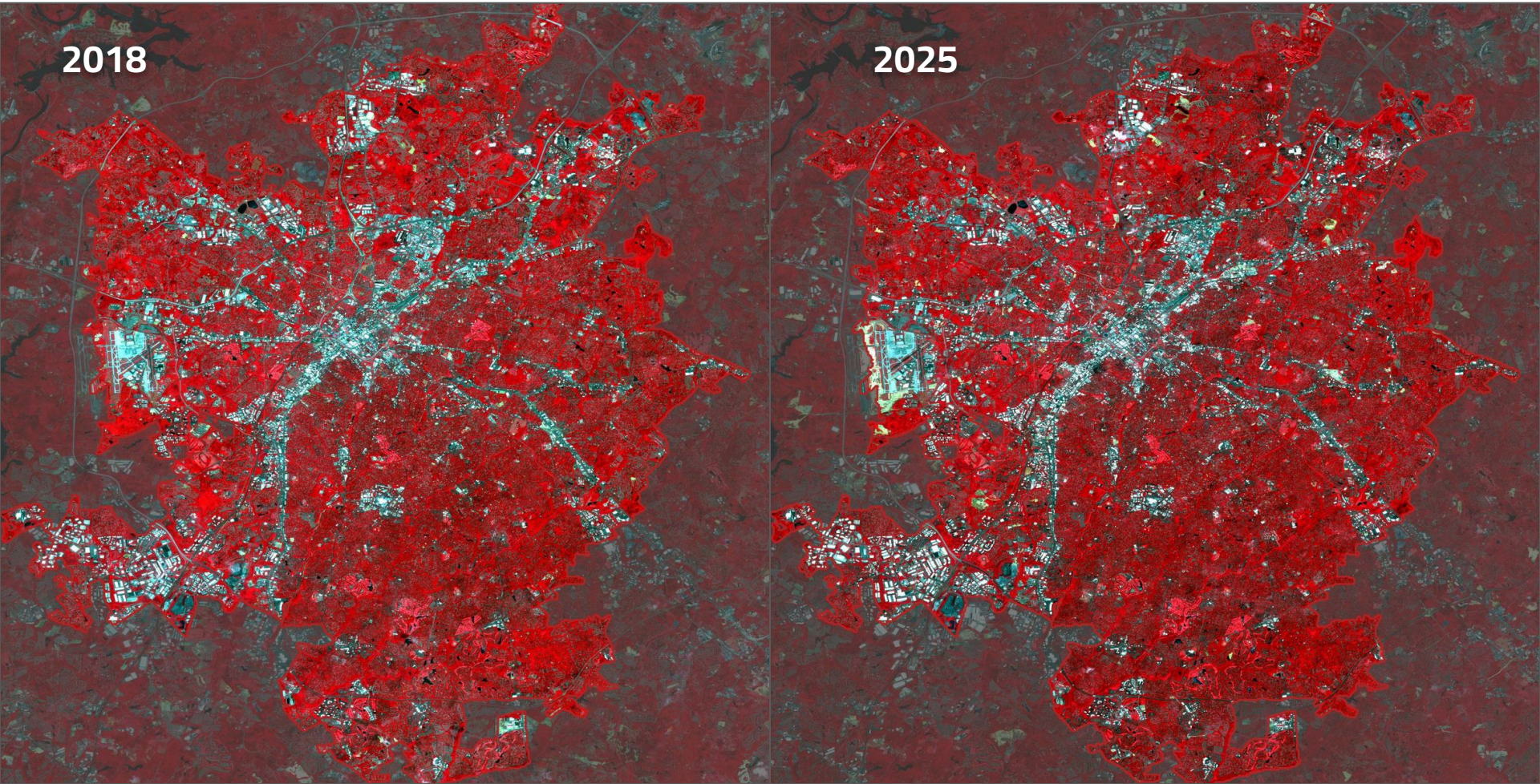
SWIR Urbanization False Color Composite *2018 vs 2025*

Bright SWIR responses highlight newly built structures and expanding development corridors



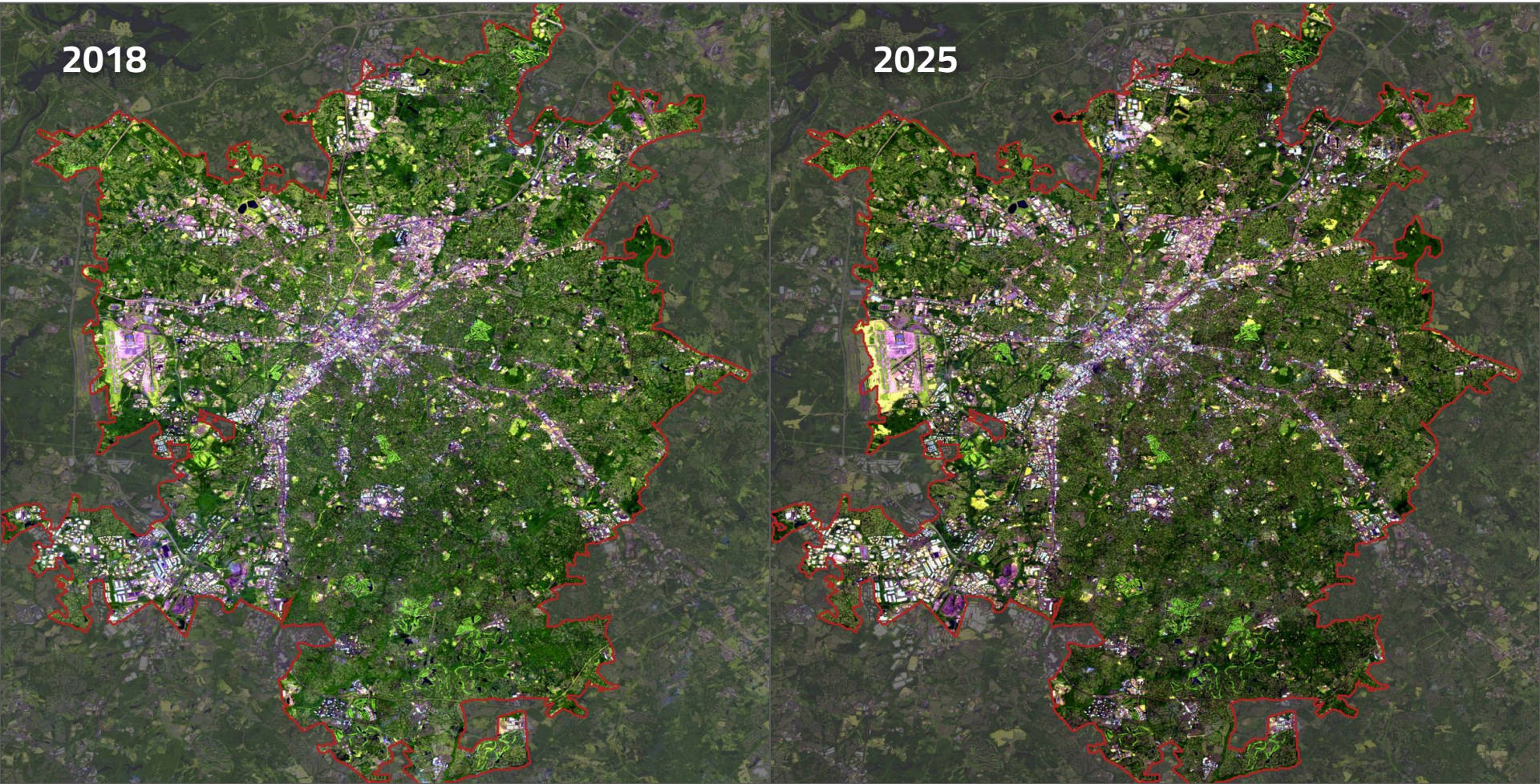
Standard Color Infrared False Color Composite *2018 vs 2025*

Fading red vegetation tones indicate widespread canopy thinning and removal across the study area



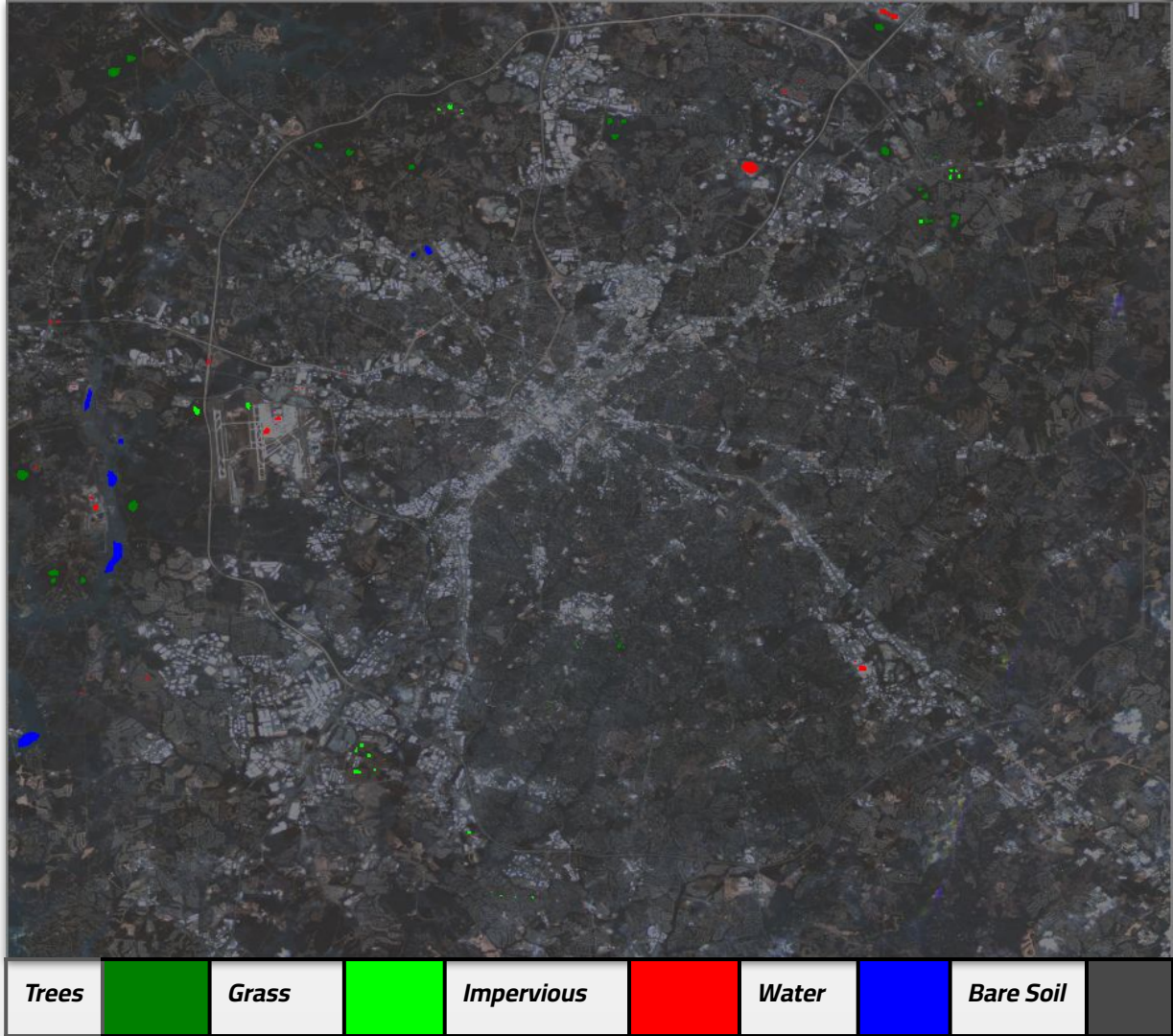
Geology and Soil False Color Composite *2018 vs 2025*

Expanding yellow soil tones reveal increased clearing, grading, and early construction activity



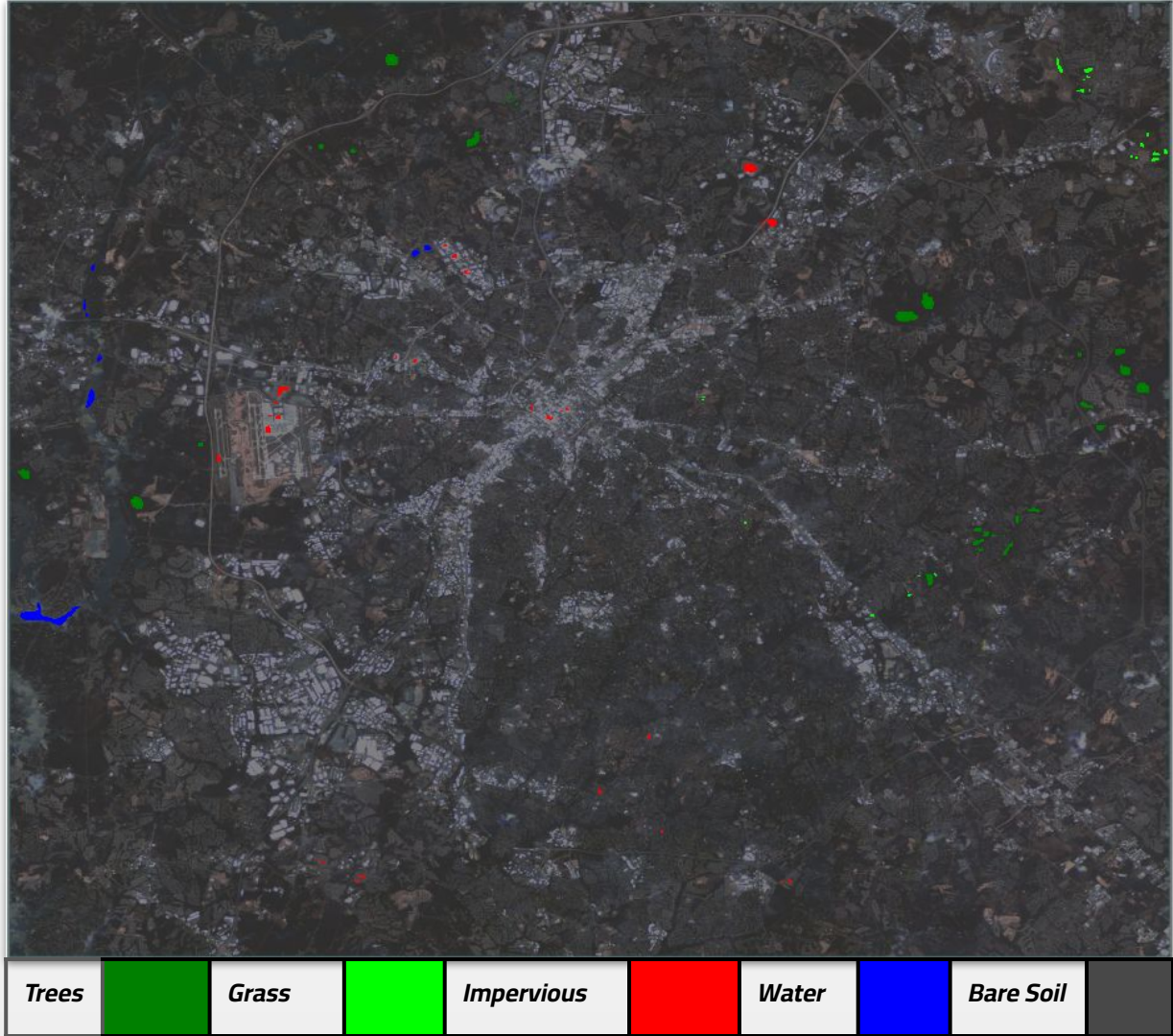
Regions of Interest for MLC Classification Charlotte NC ***2018***

ROIs capture representative spectral signatures that guide accurate classification of land cover types



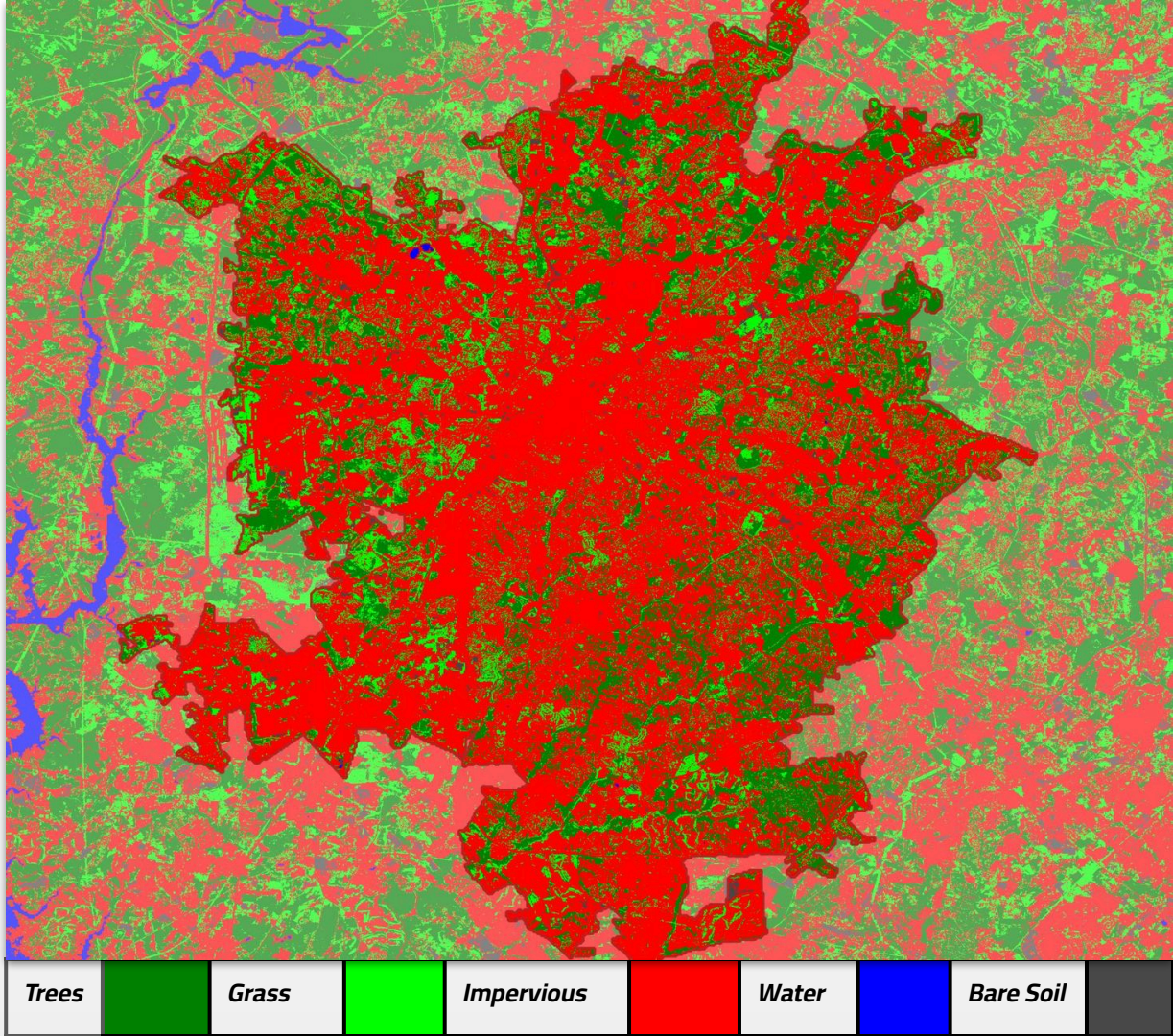
Regions of Interest for MLC Classification Charlotte NC ***2025***

ROIs capture representative spectral signatures that guide accurate classification of land cover types



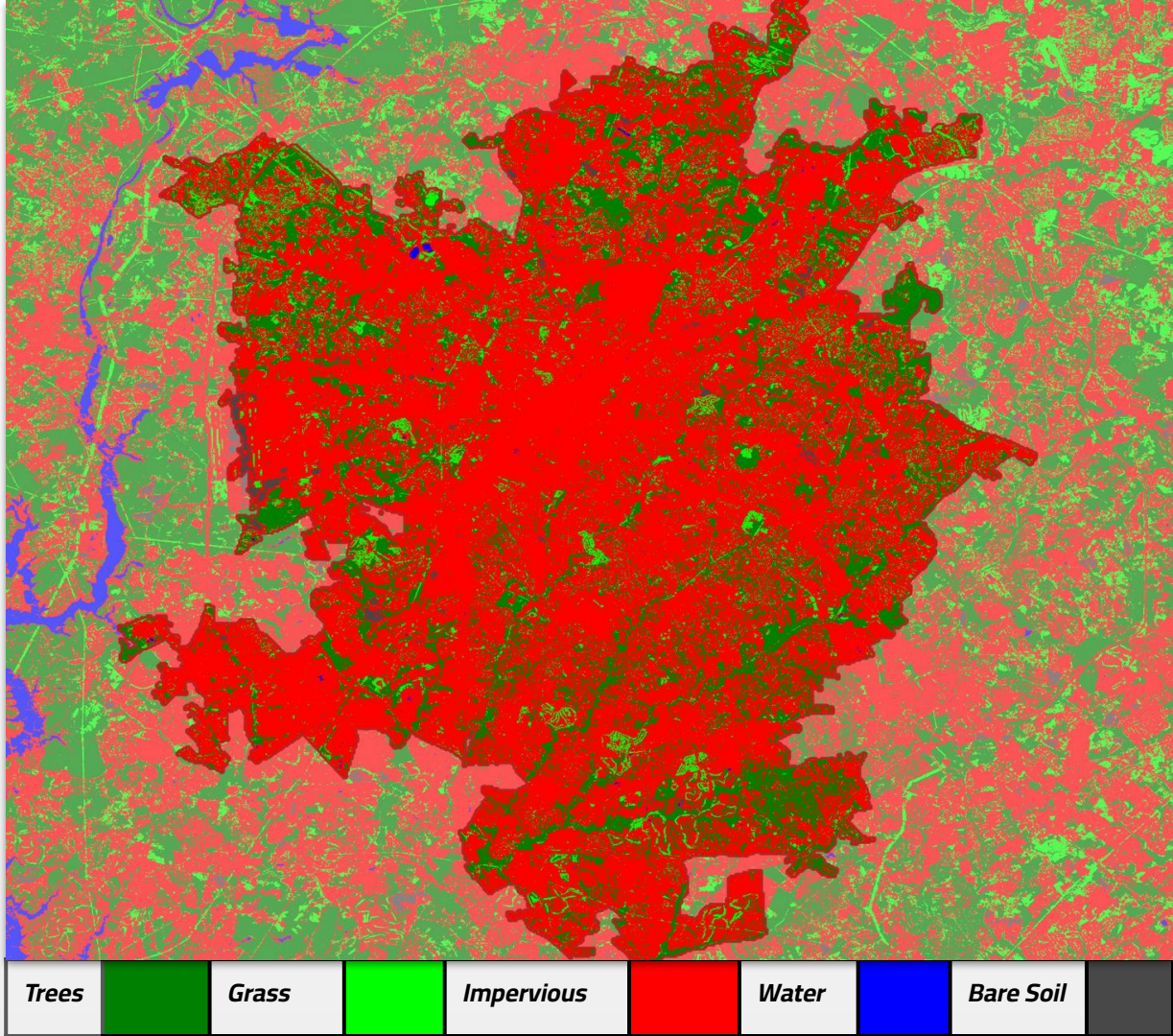
MLC
Charlotte NC
2018

Classification captures pre-development land cover dominated by vegetation



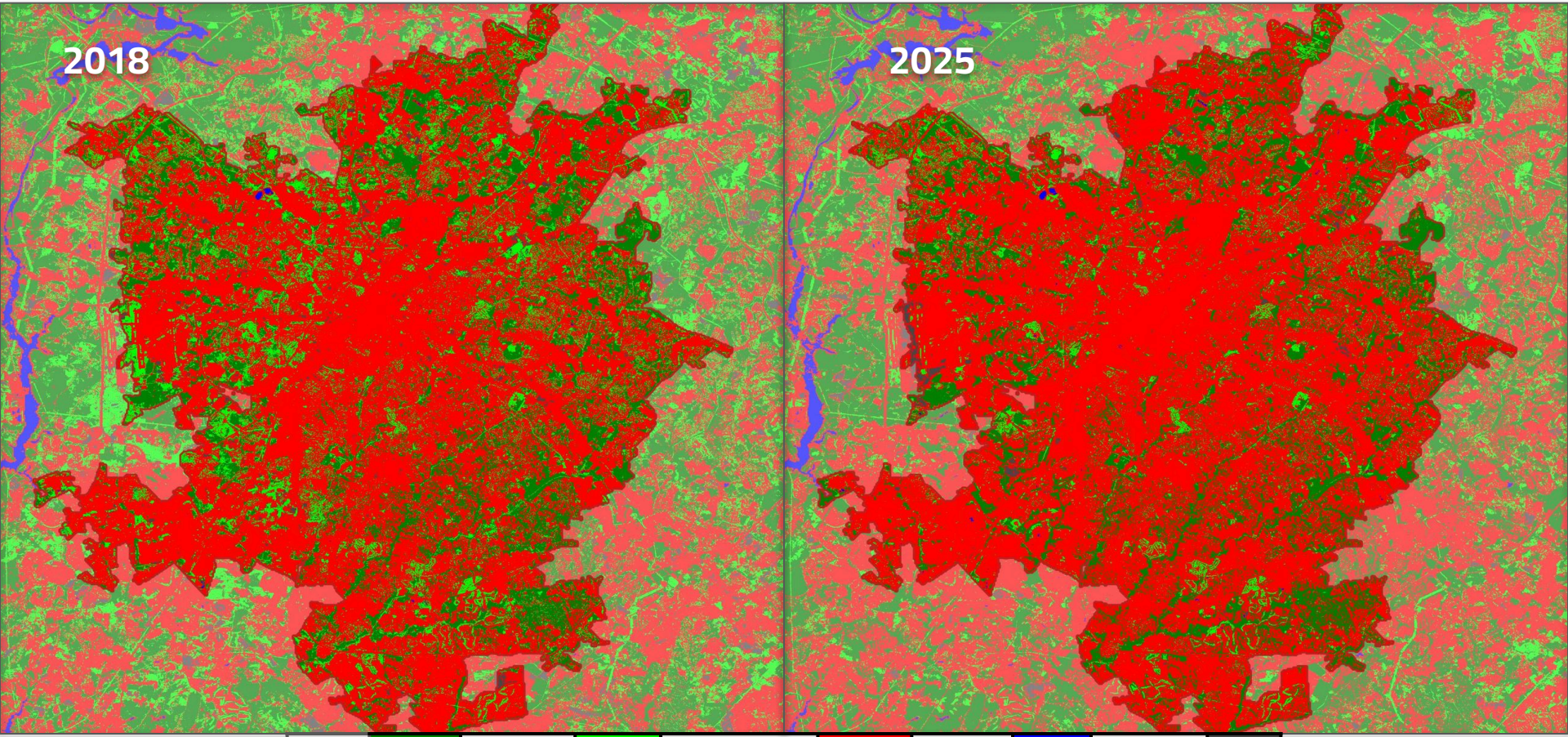
MLC
Charlotte NC
2025

Classification reflects substantial gains in urban surfaces and reductions in vegetation classes



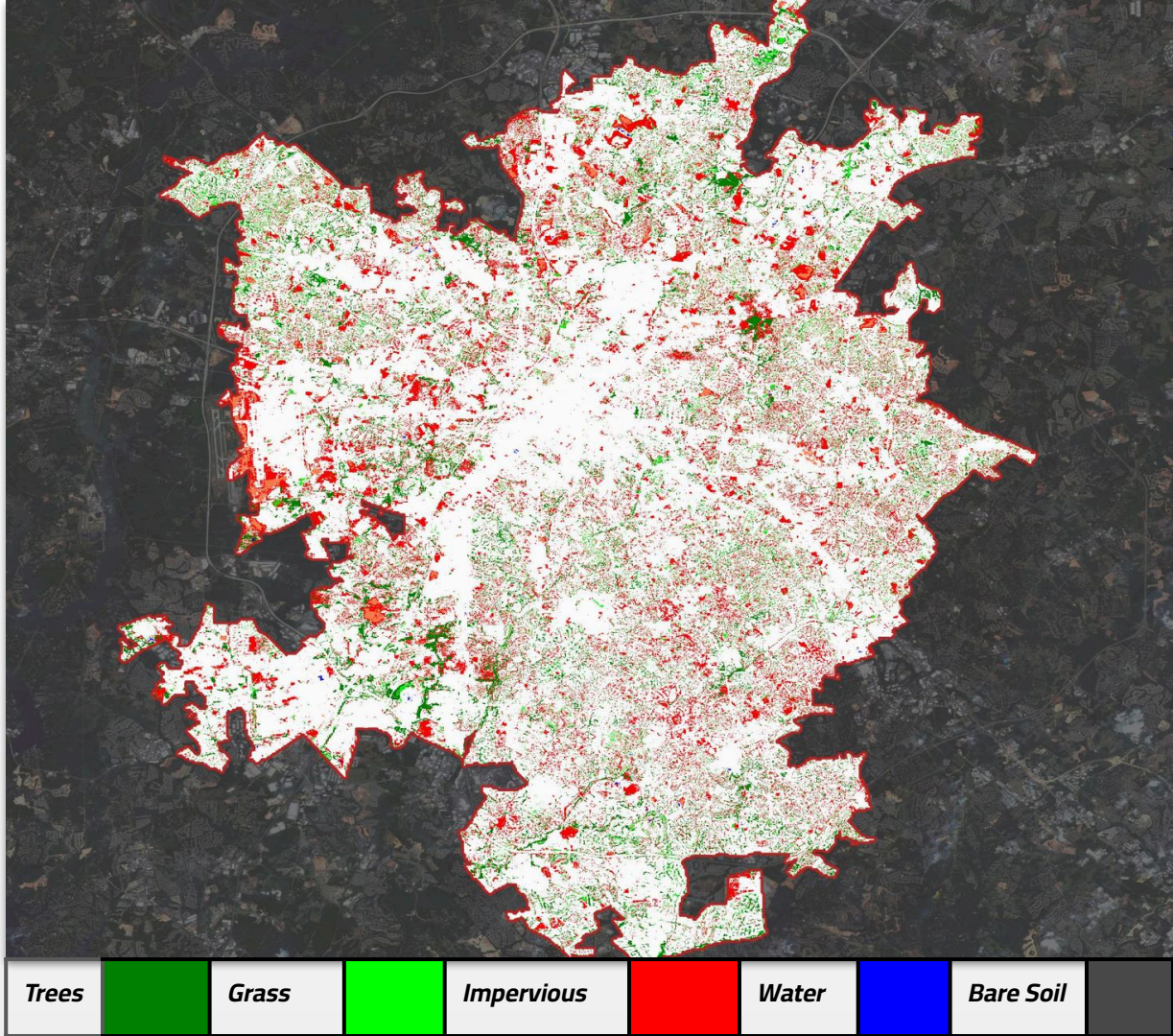
Maximum Likelihood Classification Results *2018 vs 2025*

Side-by-side classification shows vegetation converting to bare soil and impervious surfaces across the city



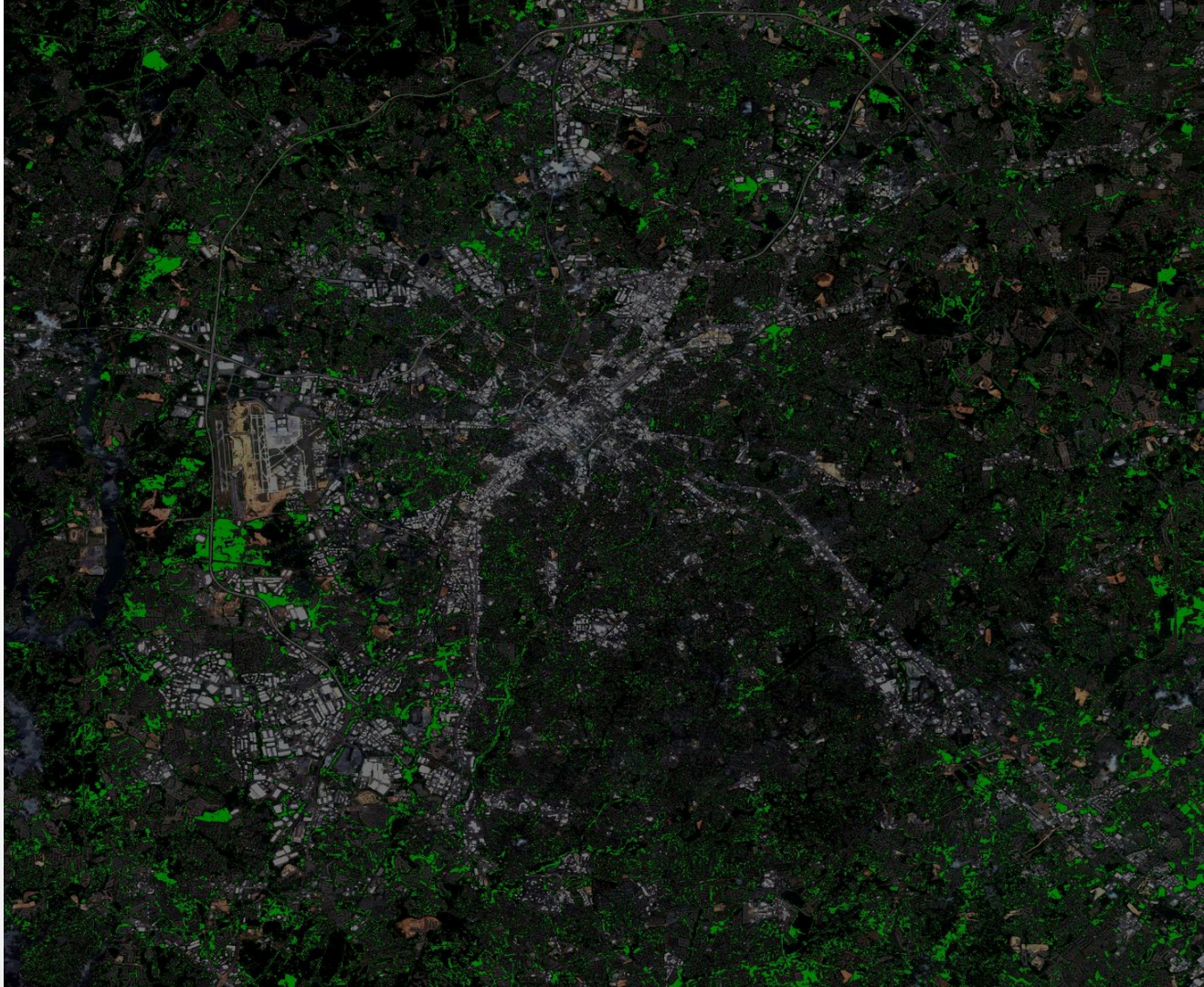
Change Detection from MLC showing *ONLY* Pixels that Changed Classes

Bright changed-pixel clusters pinpoint where land cover transitioned during Charlotte's urban growth



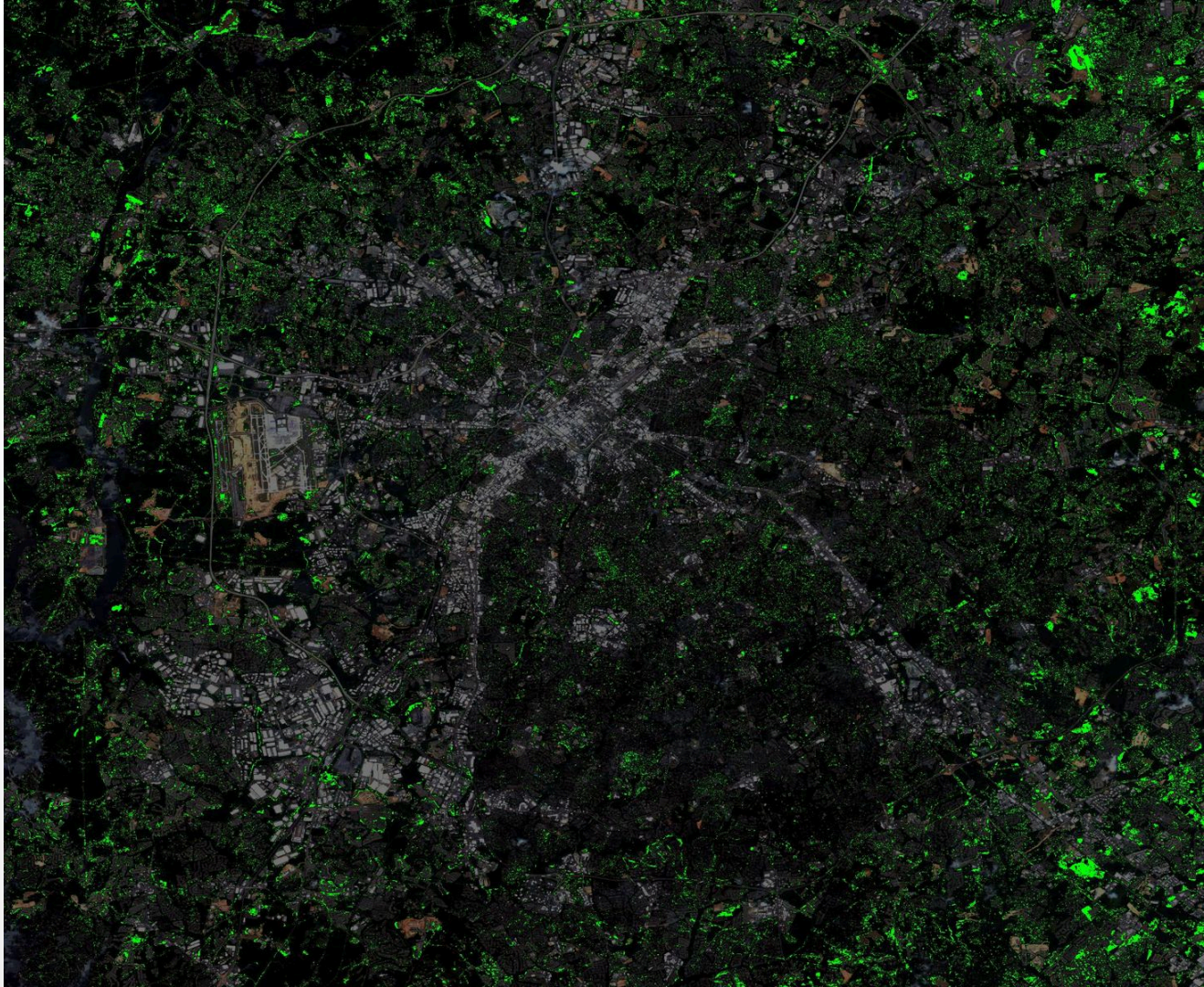
Change Detection from MLC showing *ONLY* Pixels Changed to *TREES*

Areas transitioning to tree cover reflect localized regrowth or restoration



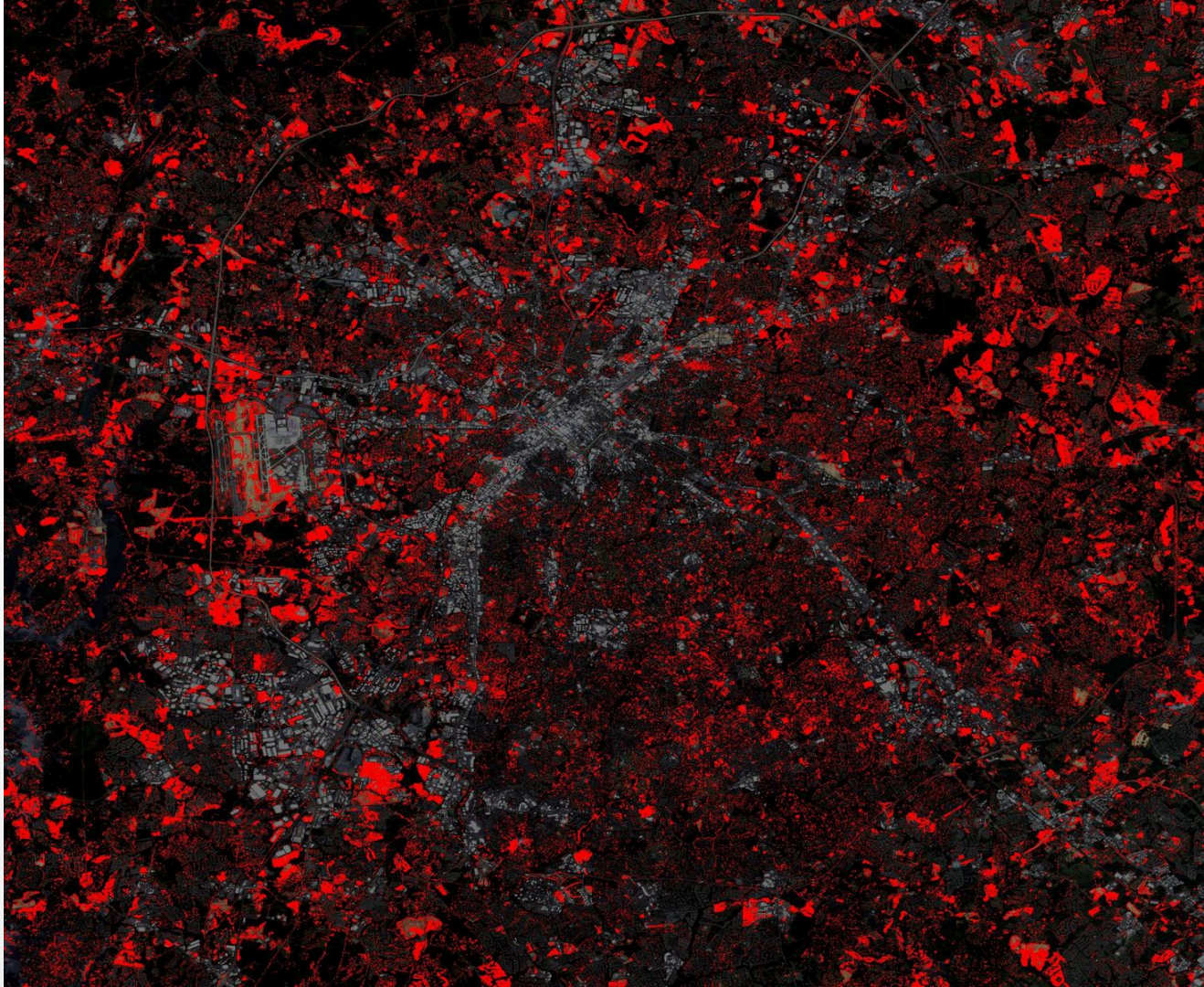
Change Detection from MLC showing *ONLY* Pixels Changed to *GRASS*

Grass gains indicate cleared vegetation or new maintained landscapes



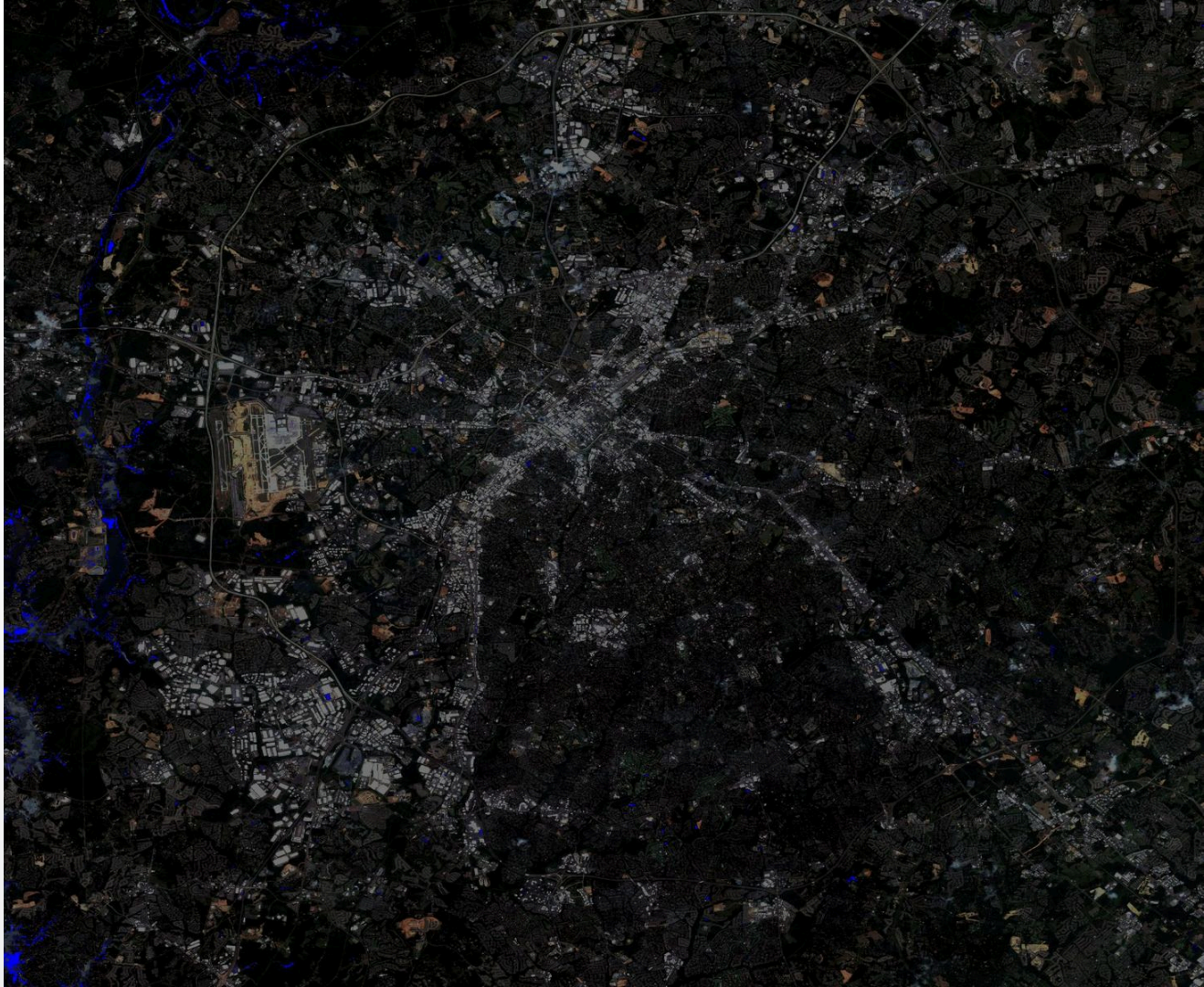
Change Detection from MLC showing *ONLY* Pixels Changed to *IMPERVIOUS*

Impervious gains identify the expansion of buildings, roads, and urban infrastructure



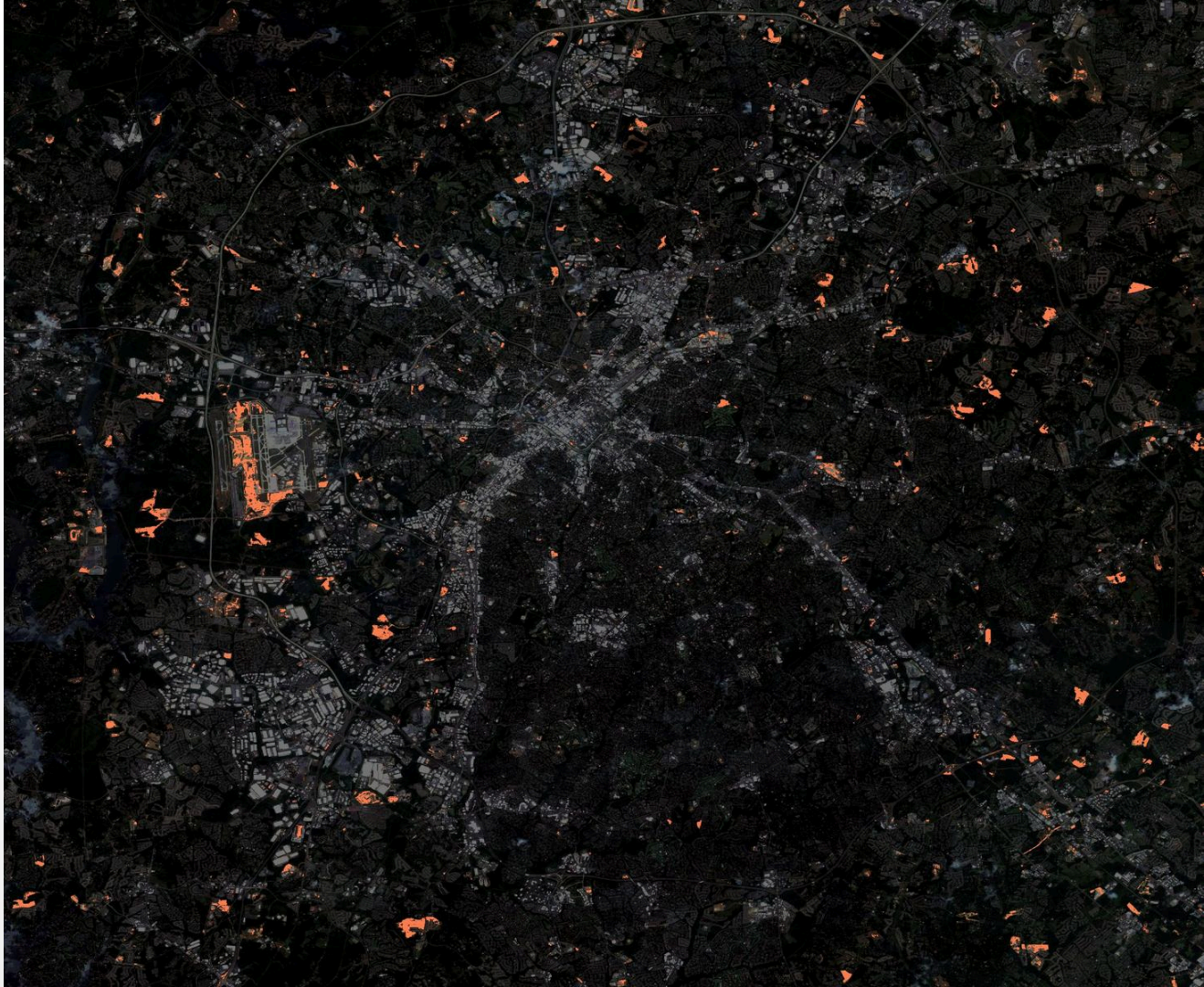
Change Detection from MLC showing *ONLY* Pixels Changed to *WATER*

New water pixels reveal added retention ponds or hydrological modifications



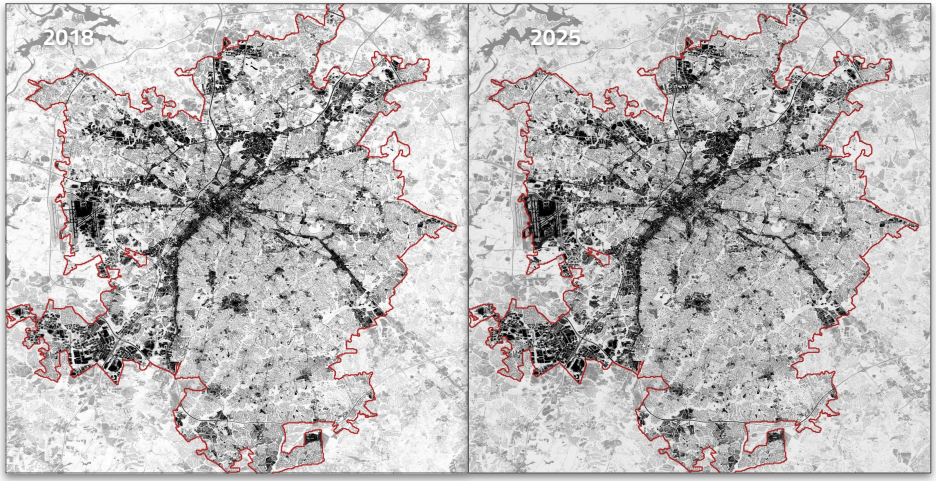
Change Detection from MLC showing *ONLY* Pixels Changed to *BARE SOIL*

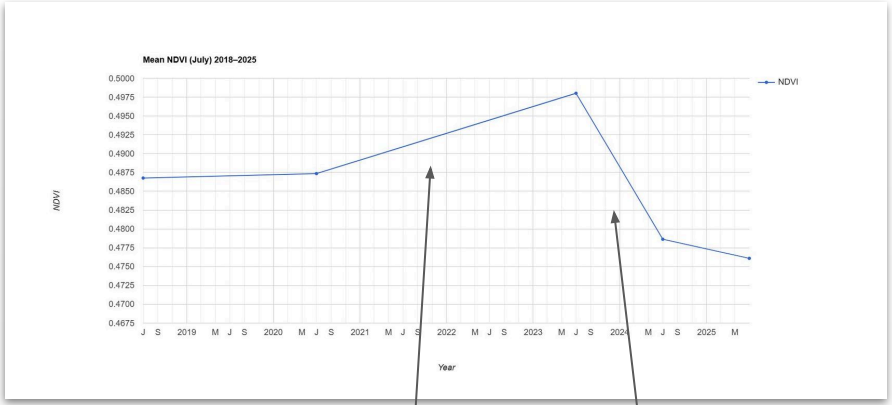
Bare soil increases highlight active construction zones and recent land clearing



ANALYSIS

NDVI and Vegetation Decline

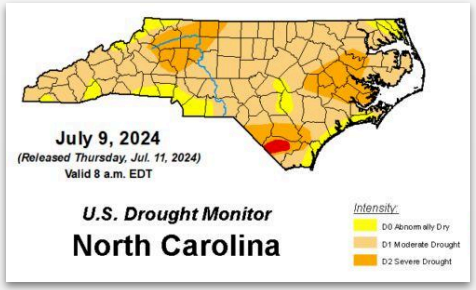




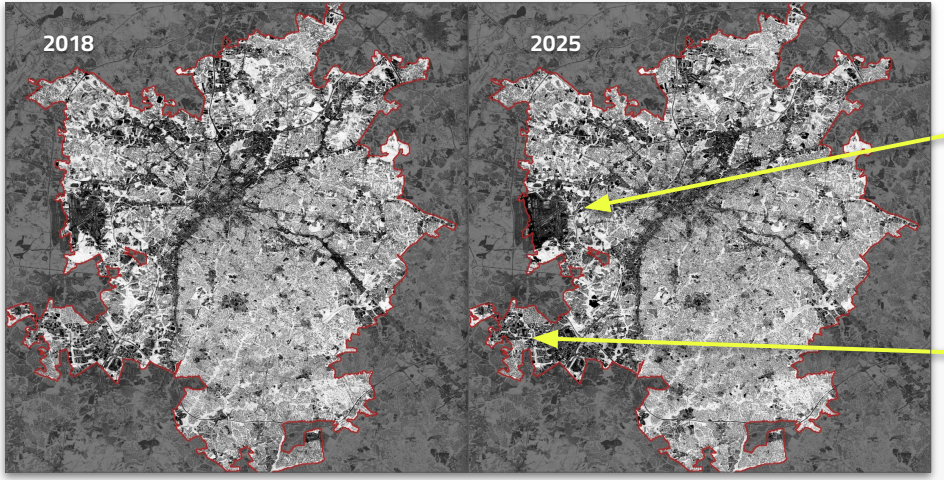
- Widespread darkening of NDVI from 2018 to 2025
    - Indicates drought confirmed through NWS data from 2024
- Large vegetation loss around suburban growth zones
- NDVI Time Series shows decline after 2022
- Aligns with Charlotte's canopy loss reports

years of high precipitation

steep decline indicative of drought / lower rainfall



Moisture and Soil Disturbance [Normalized Difference Moisture Index]

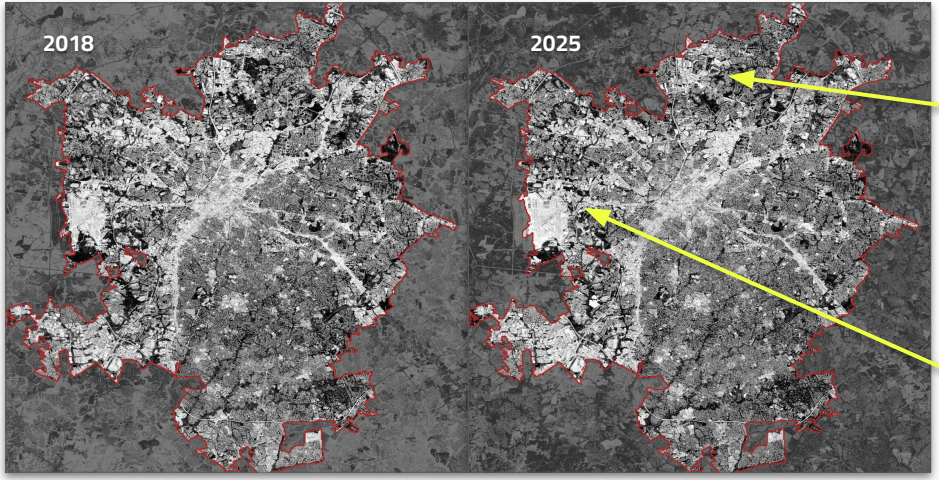


intense darkening around airport, aligns with construction of 4th parallel runway underway

darkening in southwest charlotte consistent with rapid development and growth, thus tree and vegetation loss

- NDMI values decrease overall across the study area
- Decreasing NDMI indicates drying due to canopy removal, construction and development
- Soil exposure is detected in early stage construction sites
- When matched with other spatial data, dark BLACK patches align with active development areas
- Moisture also decreased overall because of drought as discussed in the previous slide

Urban Expansion [Normalized Difference Built-Up Index]

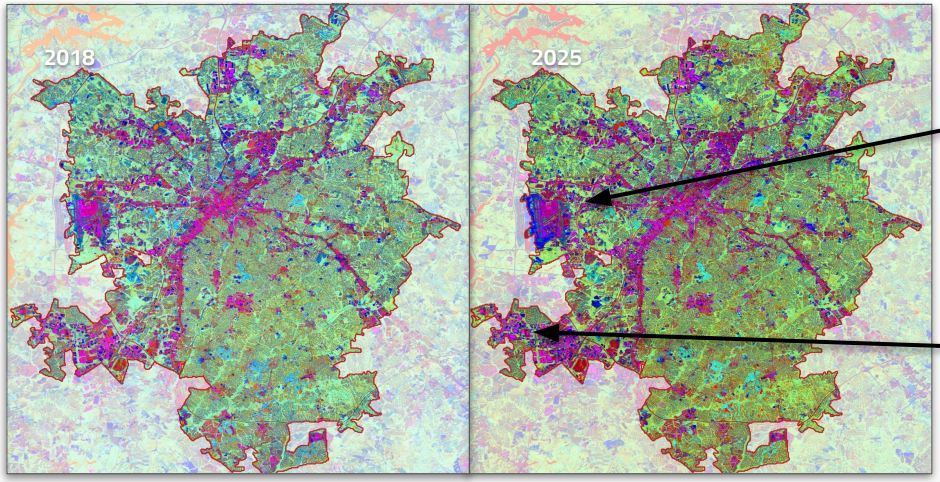


intense development in north charlotte creates brightening and shows increased impervious surfaces consistent with urban development of north charlotte and urban sprawl towards huntersville

intense brightening around airport, aligns with construction of 4th parallel runway underway and development of more gates, increasing impervious surfaces

- NDBI brightness increases significantly overall
- Built-Up Index helps highlight and track expansion and development of impervious surfaces
- Highest Urbanization in South End, Steele Creek, North Charlotte
- Matches building permit records, city development, urban growth and population trends
- Airport expansion greatly increased impervious surfaces, the airport accounts for 2.8% of Charlotte's land area

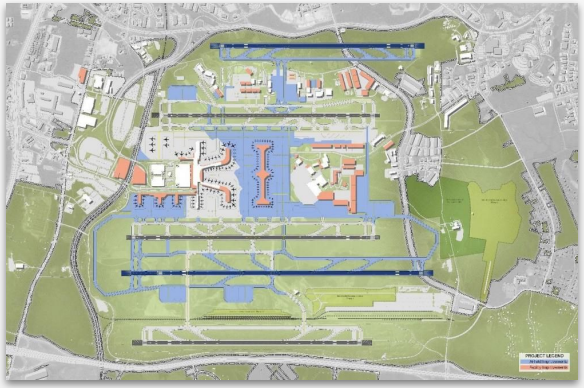
Principal Component Analysis Interpretation



drastic color change depicts spectral change as airport construction is underway losing vegetation as the land is excavated and reshaped

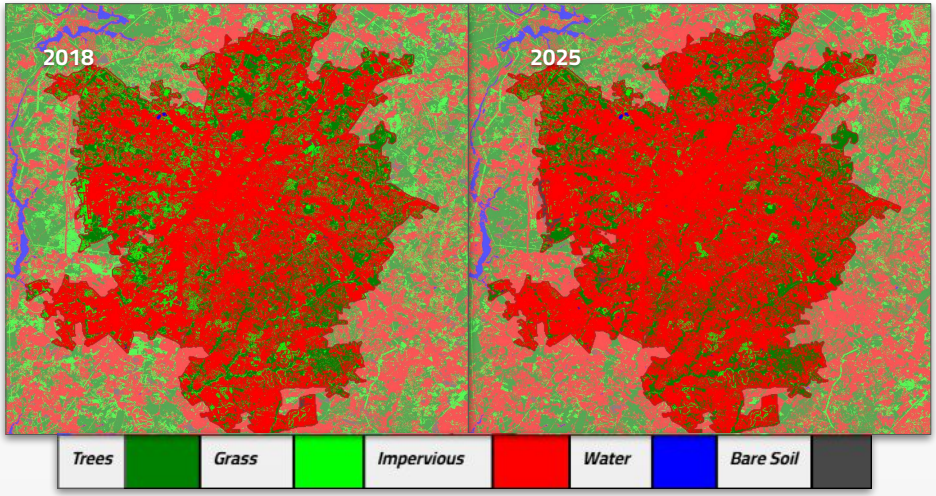
significant increase of purple/ pink tones indicating a highly developed area compared to 2018

- PCA highlights subtle spectral differences in land cover
- Increased variance in 2025 indicates landscape restructuring
- Brightened tones align with newly developed zones
- Detects early urbanization before visible completion
- Reveals spatial patterns not shown or hard to see in RGB composites

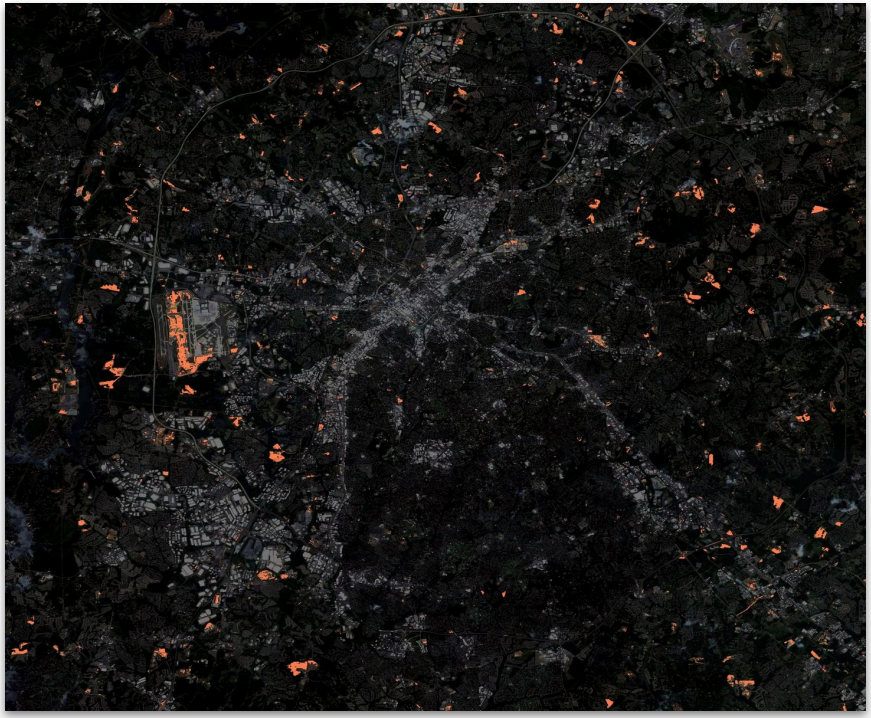


DISCUSSION

Maximum Likelihood Classification Results



- Five-Class Supervised MLC Classification
    - Trees, Grass, Impervious, Water, and Bare Soil
- 2025 shows large increase in impervious surfaces
- Vegetation classes decrease across most of Charlotte
- Change detection confirms suburban expansion patterns



the image above shows classes changing to bare soil, airport construction can be seen as well as fresh construction sites and new developments

MLC/ Change Detection Data Quality, Limitations and Correction Methods

Class Summary	Pixel Count	Percent
Unclassified	0	0.000000
Trees	1335034	7.335442
Grass	988907	5.433622
Bare Soil	119500	0.656601
Water	3010	0.016539
Impervious	5238195	28.781645

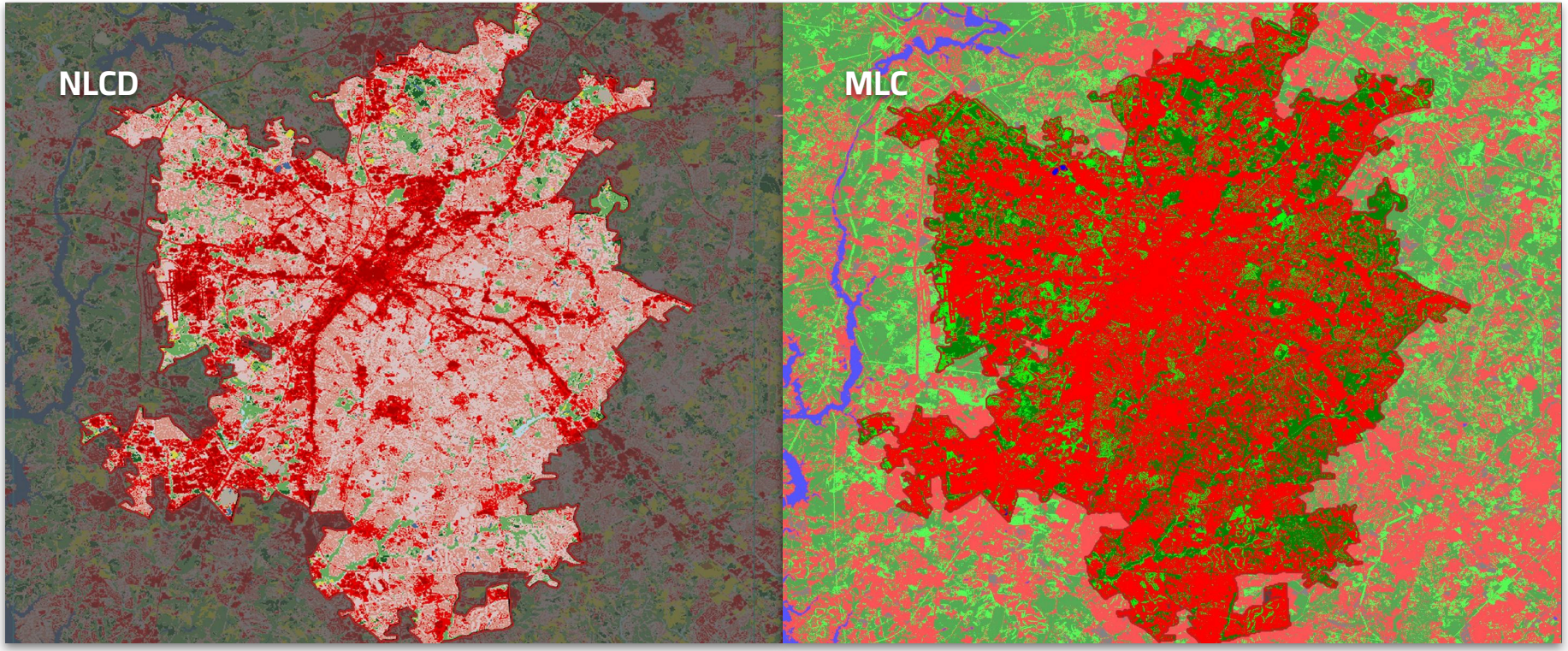
Overall Accuracy = (10348/10655) 97.1187%

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
Bare Soil	99.87	99.75	789/790	789/791
Water	98.92	100.00	5422/5481	5422/5422
Grass	100.00	85.95	1242/1242	1242/1445
Impervious	99.47	78.57	374/376	374/476
Trees	91.14	100.00	2521/2766	2521/2521

Class Summary	Pixel Count	Percent
from 'Trees' to 'Bare Soil25'	18292	0.100507
from 'Trees' to 'Impervious25'	316053	1.736576
from 'Trees' to 'Water25'	1	0.000005
from 'Trees' to 'Grass25'	32578	0.179002
from 'Trees' to 'Trees25'	968110	5.319351
from 'Grass' to 'Bare Soil25'	16786	0.092232
from 'Grass' to 'Impervious25'	500019	2.747391
from 'Grass' to 'Water25'	0	0
from 'Grass' to 'Grass25'	281845	1.548618
from 'Grass' to 'Trees25'	190257	1.045381
from 'Bare Soil' to 'Bare Soil25'	3518	0.01933
from 'Bare Soil' to 'Impervious25'	104785	0.575749
from 'Bare Soil' to 'Water25'	283	0.001555
from 'Bare Soil' to 'Grass25'	8917	0.048995
from 'Bare Soil' to 'Trees25'	1997	0.010973
from 'Water' to 'Bare Soil25'	0	0
from 'Water' to 'Impervious25'	686	0.003769
from 'Water' to 'Water25'	2324	0.012769
from 'Water' to 'Grass25'	0	0
from 'Water' to 'Trees25'	0	0
from 'Impervious' to 'Bare Soil25'	15576	0.085583
from 'Impervious' to 'Impervious25'	4783497	26.283274
from 'Impervious' to 'Water25'	3035	0.016676
from 'Impervious' to 'Grass25'	154884	0.851021
from 'Impervious' to 'Trees25'	281203	1.54509

Raw Pixel Change Numbers from Change Detection Statistics → Scale to User Accuracy from Confusion Matrix → Use Total Study Area Pixel Count → Convert Adjusted Pixel Change to Percent Increase of Study Area

MLC vs National Land Cover Database Comparison



- NLCD has different urban intensity classes while our MLC categorizes all impervious surfaces into one
- MLC is higher resolution 10 m compared to 30 m which results in sharper less smooth boundaries

MLC Conclusion and Key Findings

Accuracy-Corrected Land Cover Change in Charlotte (2018–2025)

(Clipped to official city boundary)

Class	Pixel Change (adj.)	Acres	Sq Miles	Percent Change
Vegetation (Trees+Grass)	–392,827	–9,718 ac	–15.18 sq mi	–16.9%
Impervious	+453,444	+11,209 ac	+17.51 sq mi	+8.66%
Bare Soil	–63,452	–1,568 ac	–2.45 sq mi	–53.1%
Water	+2,554	+63 ac	+0.099 sq mi	*Not meaningful*

- Our classification is decently accurate in classifying land cover types
- Confirms Rapid Expansion
- NLCD is more accurate and precise
- Trees and Grass confused

- User error and technological restrains drove our MLC to be less accurate than NLCD
- Impervious land cover was over inflated by our MLC classification
- Overall our MLC showed a decrease in vegetation consistent with NLCD

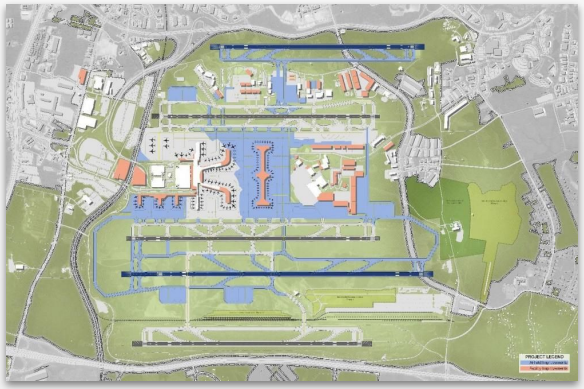
Class	Pixels	True % of Charlotte
Trees	1,335,034	17.37%
Grass	988,907	12.87%
Vegetation Total	2,323,941	30.24%
Bare Soil	119,500	1.55%
Water	3,010	0.039%
Impervious	5,238,195	68.13%

Urban Development

- As seen in previous slides, Charlotte has experienced rapid urban development
    - Charlotte has seen an increase in impervious surfaces, decreases in vegetation, and dryer soil conditions when comparing satellite data from 2018 and 2025
- NDVI analysis also show a decrease in tree canopy
- SWIR composites further highlight the increase in impervious surfaces in Charlotte, as well as the dryer conditions found in the study area
- Due to rapid vegetation loss and dramatic increase in impervious surfaces, runoff water management may be an issue in the future, potentially exacerbated by continuous development

Vegetation Changes

- As observed in the MLC images, between 2018 and 2025 Charlotte saw a large decrease in vegetated areas, as they were converted into impervious surfaces
- Monitoring vegetation changes, especially in cases where trees or grassy areas are converted into bare soil, can be used to infer where construction projects are likely to happen in the future
    - In the MLC a large area of bare soil can be observed near Charlotte's western border, due to its proximity to Charlotte-Douglas International airport, this area has most likely been cleared to be built upon as an addition to the airport



Population Growth

- Charlotte is one on the fastest growing cities with an annual growth rate of around 1.8%. As population increases, so will impervious surfaces
- As large scale construction projects continue, the trend of decreased vegetation and increased impervious surfaces will also continue
    - In the western part of Charlotte the River District can be found. This is a 1200 acre project zoned as multi-use, but has a focus on residential spaces
    - The project has listed one of its goals as preserving nature, and offers green spaces and hiking trails. Even so its site can be clearly seen in the images, most noticeably the RGB composite
    - The western side of Charlotte seems to be an area with alot of impervious surfaces, and bare soils indicating future construction plans. This is likely due to the location being near the airport and several major highways.

CONCLUSION

Conclusion

- When examining satellite data of Charlotte, NC between the years of 2018 and 2025 it is clear that the city experienced rapid urban development
- This development can be viewed as vegetation is transformed into bare soil, and then into a built (often impervious) surface
    - 2025 imagery shows large decreases in tree canopy cover, and increases in bare soil coverage indicating this growth pattern will continue at perhaps an even greater rate as the cleared area is constructed upon

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QUESTIONS?